Technical Documentation

for

NU Laguna Queuing System

Version 1.0 approved

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NU Laguna

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1. INTRODUCTION

1.1 Purpose

At NU Laguna, students often experience long waits during important administrative tasks like enrollment, paying fees, and getting important documents at the registrar office. These delays results stress and time-consuming for the students and for the staff.

To solve this problem, we propose creating a queuing management system to make everything run more smoothly. The Queuing Management System simplify the flow of the students in queues and effectively reduce waiting times.

The system will also direct students to the next available service window, making sure no window is left unused. It will track data on arrival times, wait times, and service times to help improve the process. By looking at this data, the system can identify busy times and suggest how many staff members are needed, ensuring there are enough people to help during peak times.

By using Queuing Management System, it will reduce the waiting times for students and make the administrative process more efficient and less stressful. This system will help both students and staff to be more organized and better overall experience.

1.2 Document Conventions

This document uses the following conventions:

Abbreviations	**Meaning**
VS Code	Visual Studio Code
XAMPP	XAMPP Apache MySQL MariaDB PHP Project
PHP	Hypertext Preprocessor
SQL	Structured Query Language
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
AJAX	Asynchronous JavaScript and XML

1.3 Requirements

- **Software Requirements:**
    - XAMPP version 8.2.12 for windows 10
    - Visual Studio Code for windows 10
    - HTML
    - Bootstrap

- CSS
- Java Script
- AJAX
- PHP
- SQL
- SweetAlert2
- JQuery

- **Operating System**
  - Windows 10 Home 64-bit
  - Windows 11

- **Development Tools and IDE**
  - Visual Studio Code
  - XAMPP
  - PHP
  - Canva
  - SQL
  - HTML
  - Bootstrap
  - CSS
  - AJAX
  - JQuery
  - SweetAlert2

- **Hardware Requirements**

Central Processing Unit	AMD Ryzen 3 3200G
Motherboard	ASRock B450M-HDV R4.0
Random Access Memory	8GB DDR4 Kingston Hyper Fury
Storage	240GB SSD Kingston A400
Power Supply Unit	450W PSU EVGA 450 BT

1.4 System Architecture

Frontend (Client-side):

- **Technologies:**
  - HTML
  - Bootstrap
  - CSS

- JavaScript

- **Description:**

- The frontend is built using HTML for the structure, CSS and bootstrap for styling and JavaScript for dynamic interactions. HTML provides the foundation of the web pages, CSS style for the content, Bootstrap ensures responsive design, and JavaScript adds interactivity.

- **Key features:**

- **Responsive Design:** Bootstrap ensures that the application is accessible and looks good on a variety of devices.
- **Dynamic Interactions**: JavaScript enables interactive elements such as form validations, dynamic content updates, and AJAX requests for communicating with the backend without reloading the page.
- **Static Assets Management:** CSS and Bootstrap manage the visual styling and layout, while JavaScript handles client-side logic and user interactions.

Backend (Server-side):

- **Technologies:**

- jQuery
- SweetAlert2
- Bootstrap

Description:

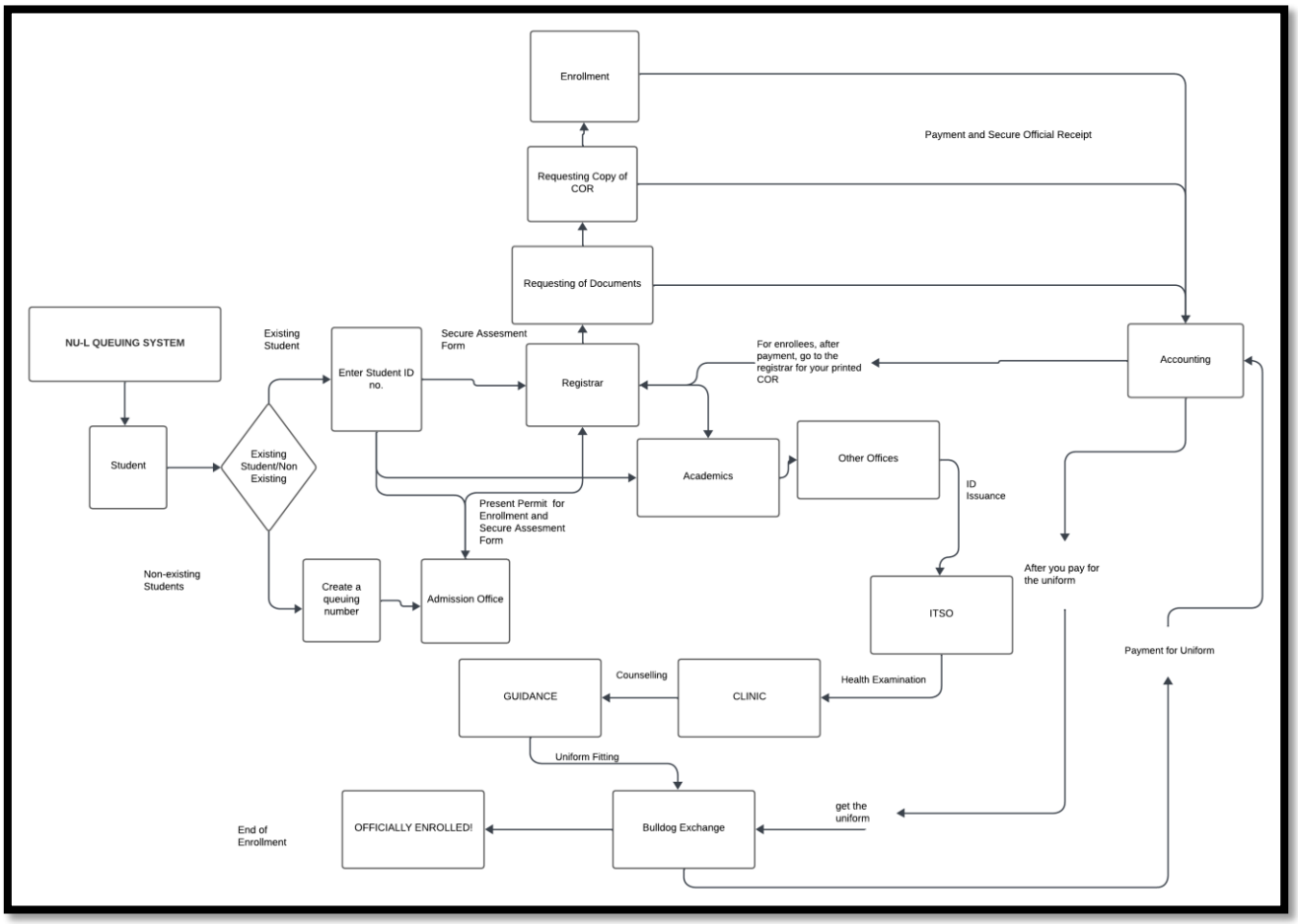
- jQuery is a small fast, and feature-rich JavaScript library; SweetAlert2 improves and customizes alert messages; and Bootstrap includes CSS- and JavaScript-based design templates for forms, buttons, navigation, and other interface components.

Key Features:

- **DOM Manipulation**: Simplifies HTML document traversal and manipulation, making it easy to select, modify, and traverse DOM elements.
- **AJAX Support**: Simplifies asynchronous HTTP requests, allowing for easy retrieval and sending of data without refreshing the page.
- **Beautiful Alerts**: Creates customizable, attractive alert dialogs that can replace default browser alerts.
- **Themes and Styling**: Offers different themes and custom styling options to match the look and feel of your application.
- **Responsive Design**: Provides a grid system and responsive classes to create layouts that adapt to different screen sizes and devices.

- **Utility Classes**: Includes a large set of utility classes to control spacing, alignment, colors, and other aspects of design.
- **Consistent Design**: Ensures a consistent look and feel across different parts of a web application.

1.5 System Flowchart



2. OFFICES AND MODULES

2.1 SUPER ADMIN

2.1.1 Overview

The super admin module, which will be used by ITSO, plays the most important role in the NU Laguna queuing system due to its functionalities, which are crucial for the other offices.

In this module, the super admin has administrative functions like managing records and reports for data analysis for each office. It also has the functionality of adding, updating, and deleting an office, demonstrating its scalability when new offices are added to the campus. This module can export data in PDF and XLS file formats to ensure data accuracy.

Furthermore, the module can manage users depending on the offices to which they are assigned. The super admin can also manage the programs and academic strands, to accommodate the changes in the university's programs and/or strands.

In conclusion, the super admin module is essential for maintaining the efficiency and scalability of the NU Laguna queuing system, ensuring accurate data management and seamless administrative operations across the campus.

2.1.2 Functionalities & Screenshots

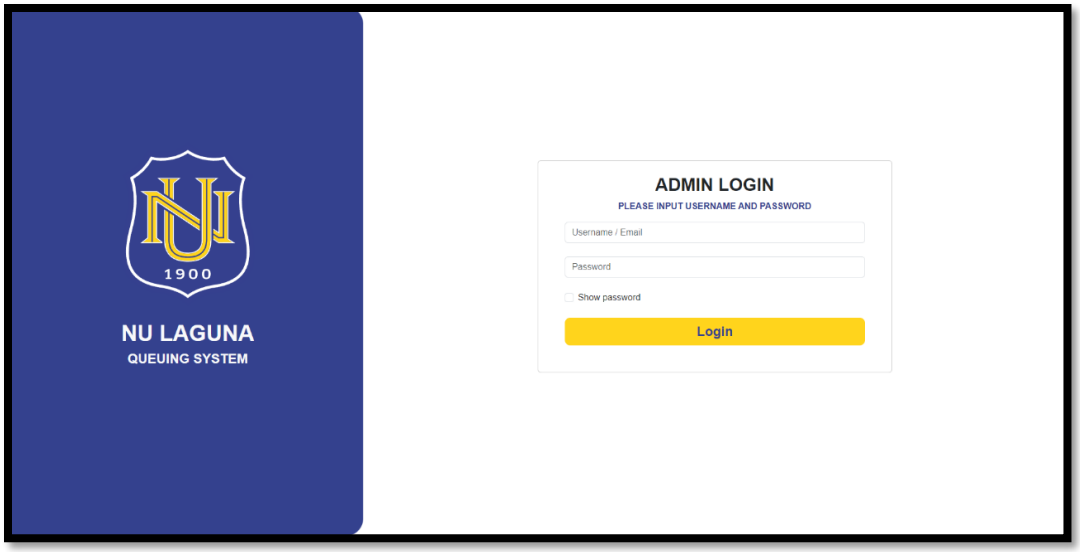
- **Login Interface**
- The login interface provides a way for super administrators to access the super admin module.

Features:

- Username Input Field
- Password Input Field

Actions:

- If the user inputs the correct credentials, it will redirect them to the super admin dashboard.
- If the user inputs incorrect credentials, an error displaying “Incorrect email or password” will appear.
- If the user does not input an email or password, a pop-up will appear that will prompt the user to fill out the field.



- **Super Admin Dashboard**

The super admin dashboard displays data that the super admin needs to monitor traffic and overlook the queues and offices.

Features:

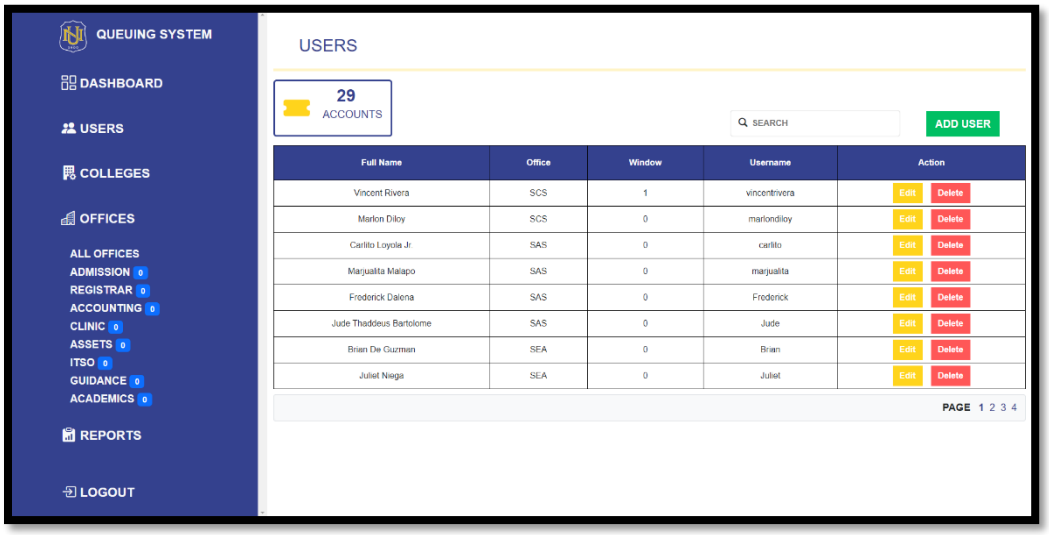
- Number of Customers, Completed, and Pending
- Graph of average transaction time per office
- Graph of current number of customers

- **Users Dashboard**

The user dashboard displays all the accounts for all the different offices. The super admin can use the user dashboard to add, edit, search, and delete accounts.

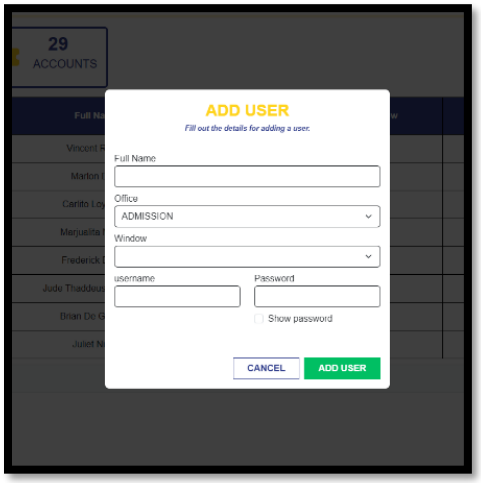
Features:

- Add User
- Edit User
- Display User
- Search User
- Total number of users

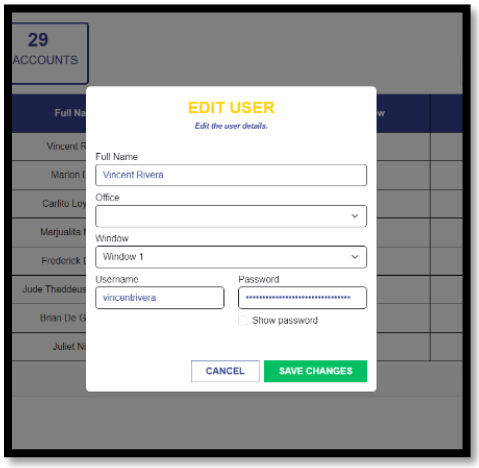


Actions:

- When the "ADD USER" button is clicked, a modal will appear wherein the super admin can input the Full Name, Office, Window, Username, and Password.



- When the “EDIT” button is clicked, a modal will appear wherein the super admin can edit the Full Name, Office, Window, Username, and Password of the selected user.



- When the “DELETE” button is clicked, a pop-up will appear to confirm if the super admin wants to delete the selected user.
- The user can use the search bar to find the user, which will be helpful when the number of users too many.

- **Colleges Dashboard**

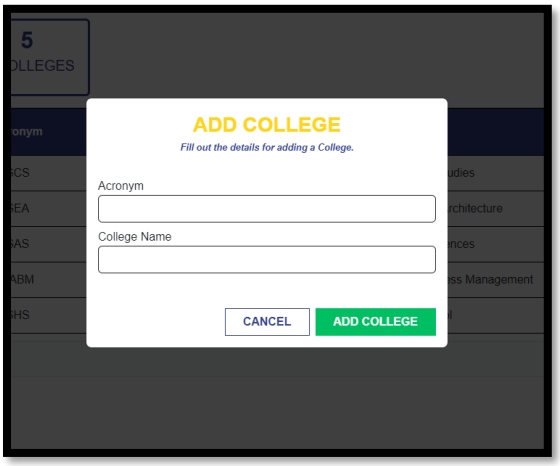
The colleges tab displays all the colleges in NU Laguna and allows the super admin to manage them. The user can search, add, edit, and delete a college to adapt to the changes in the university.

Features:

- Total number of existing colleges
- Search College
- Add College
- Edit College
- Delete College

Actions:

- The user can search for the name or acronym of the college using the search feature, making it easier to find the college, especially when the list becomes too long.
- Clicking the "ADD COLLEGE" button will trigger a modal where the user can assign an acronym and input the college's name.



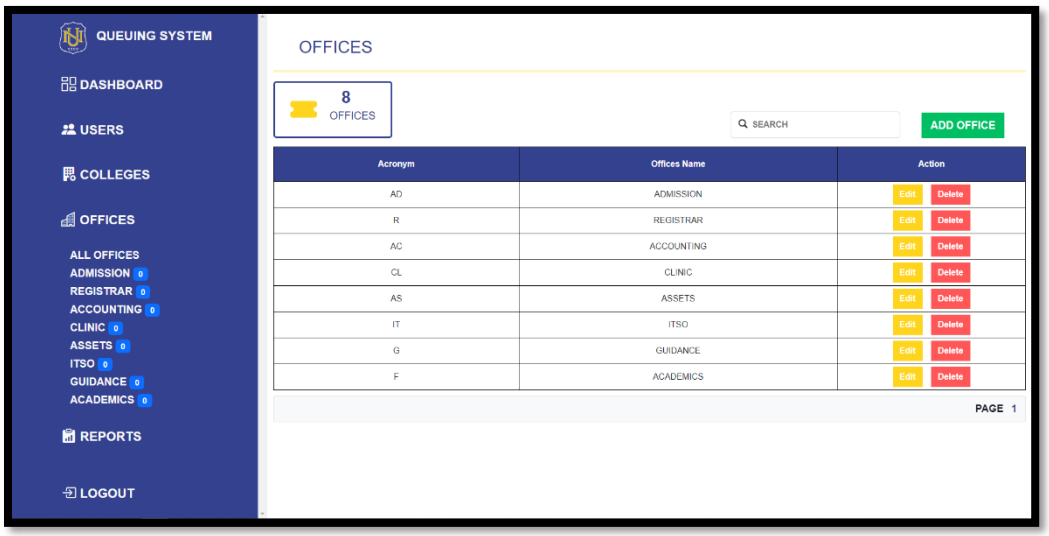
- When the user clicks the "EDIT" button, a modal will pop up to edit the acronym and name of the college.
- If the user clicks the "DELETE" button, a message will pop up to confirm the action.

- **Offices Dashboard**

The Offices dashboard enables the super admin to oversee and manage the list of offices included in the queuing system.

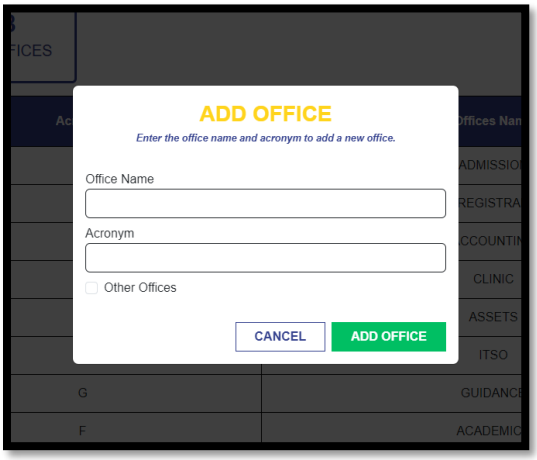
Features:

- Total number of existing offices
- Search Office
- Add Office
- Edit Office
- Delete Office



Actions:

- The user can search for the name or acronym of the office using the search feature, making it easier to find the office, especially when the list becomes too long.
- Clicking the "ADD OFFICE" button will trigger a modal where the user can assign an acronym and input the office's name.



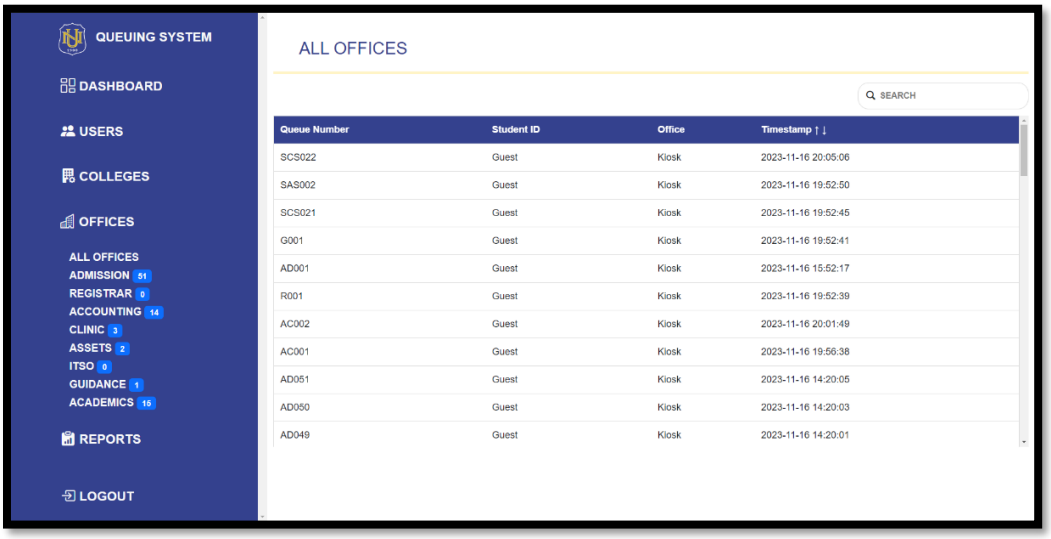
- When the user clicks the “EDIT” button, a modal will pop up to edit the acronym and name of the office.
- If the user clicks the “DELETE” button, a message will pop up to confirm the action.

- **Offices Queues**

The All-offices module consists of all queues on each office, it also consists of each office to monitor the status of the queue numbers.

Features:

- Search
- View queue number details



Actions:

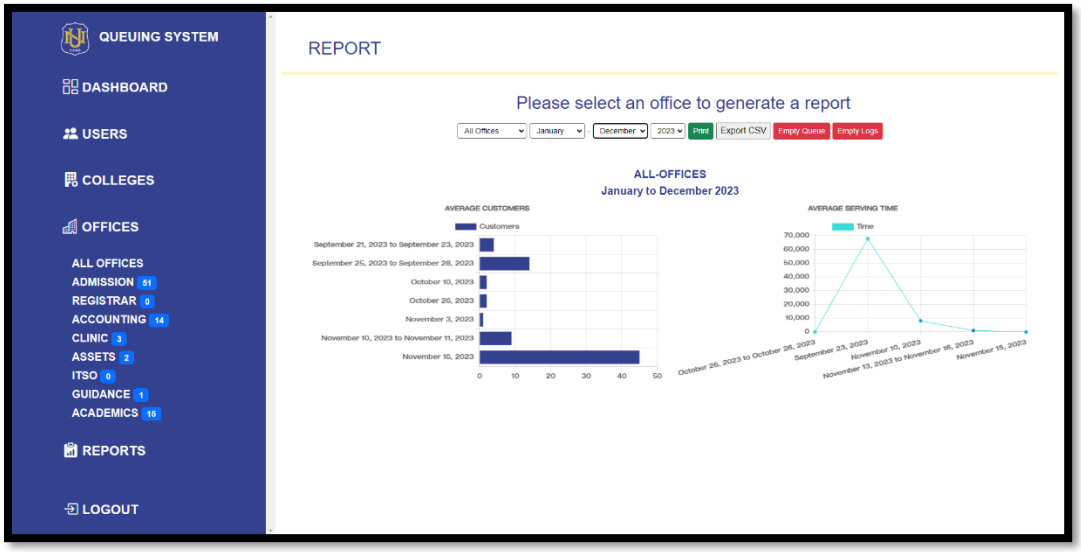
- The user can use the search bar to find a specific queue number. Then, only the data that matches the search is displayed on the table.
- When the user clicks a specific row, it will display the details of every transaction of that number.

- **Reports Dashboard**

The Reports Dashboard module consists of reports that can be generated by selecting the office, month, year, and several actions to make a report. The reports can be exported to CSV or PDF format. It allows the administrator to generate reports to monitor the number of queues that the NU Laguna has.

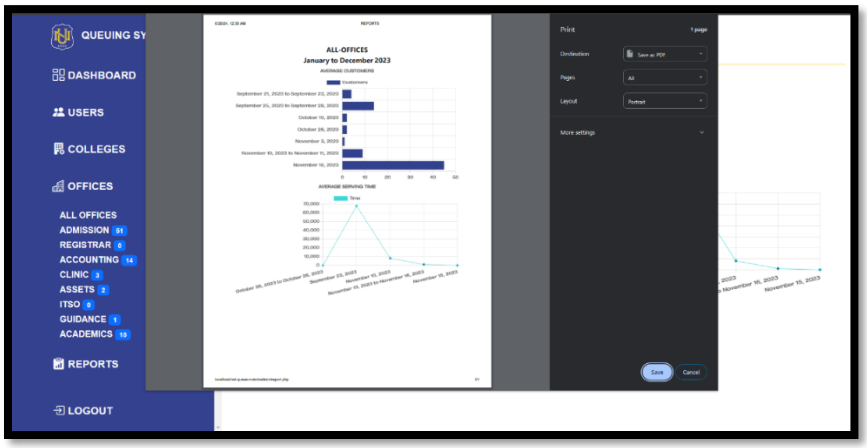
Features:

- Office filter
- Date filter (range)
- Print reports
- Empty Logs
- Empty Queue



Actions:

- If the user wants to generate a report, they can choose the desired office by clicking the first drop down button.
- The user can choose the date range (month & year) from the drop-down button and the user would easily see the generated report or the number of queues.
- By clicking "Print" it will give the user an option to print or save as PDF
- By clicking "Export to CSV", the generated report will be downloaded as CSV –
- When the "Empty Logs" button is clicked, the system will clear all the logs or the completed queues at once.
- When the "Empty Queue" button is clicked, it will clear all the existing queues.

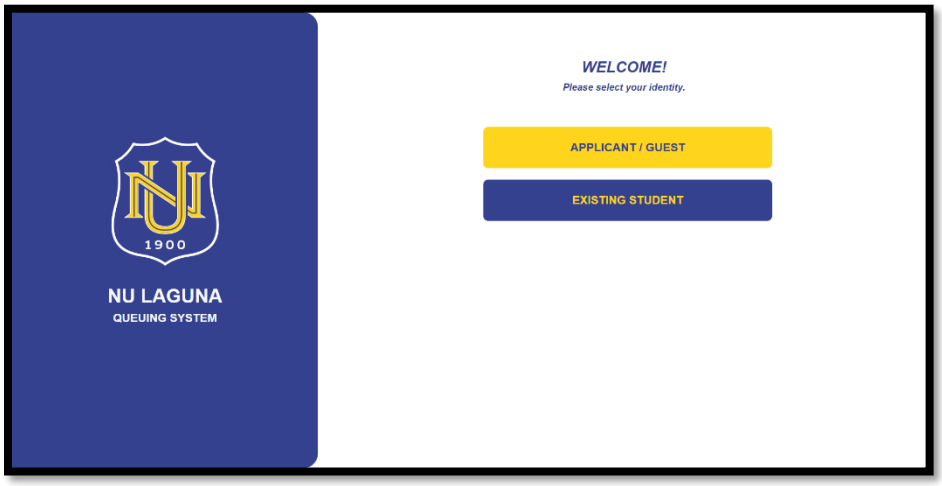


- **Student / User Interface**

The student / user interface allows existing students, applicants, and guests to generate a queue number for their desired office.

Features:

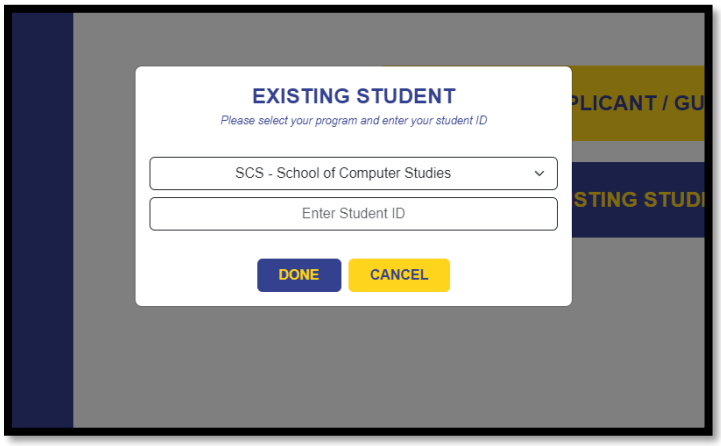
- Identity selection
- Existing student log in form



Actions:

- A guest or applicant may queue for their selected office by simply clicking "Applicant / Guest" button.

- For existing students, they must click the “Existing Student” button for them to log in and validate their identity. A pop-up will appear wherein existing students may select their program and their student ID.

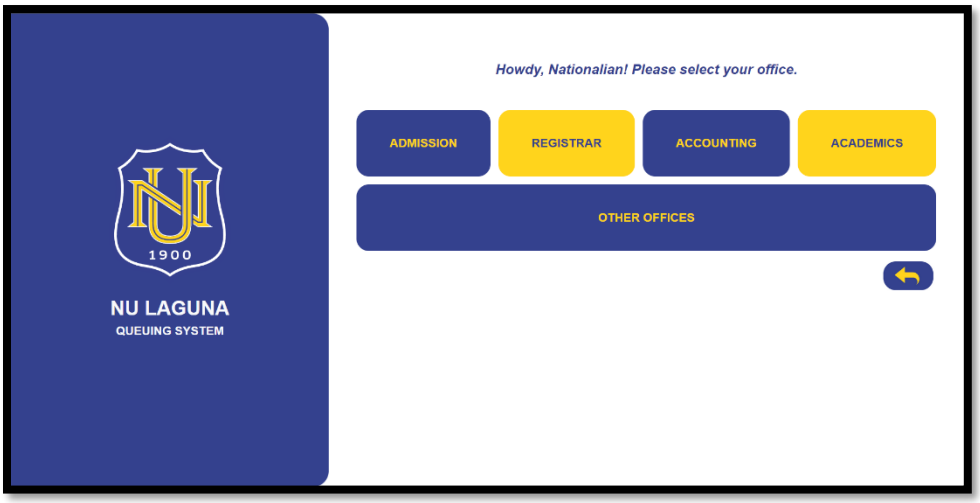


- **Offices Interface**

The offices interface displays all the existing offices. This is where students can select and queue up for their chosen office.

Features:

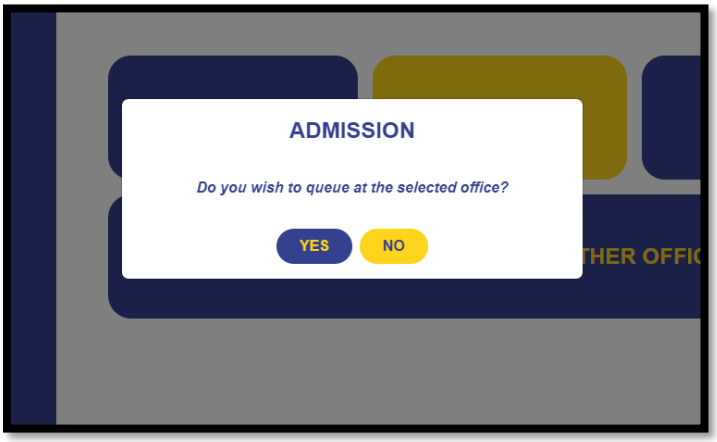
- View offices
- Queue to offices.



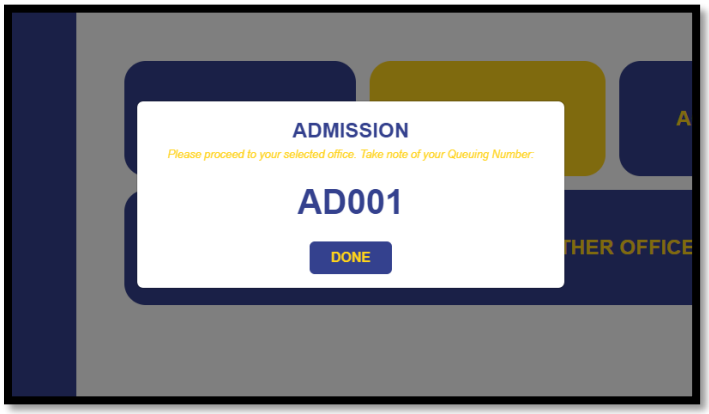
Actions:

- Students can use the offices interface to select and queue up at the selected office.

- To queue up for an office, the user can click on the office button and a pop-up will appear asking for confirmation.



- When the user clicks "yes", it will automatically display and print their queue number. If the user clicks "no" the pop-up will close.



- **Main interface/Display queue**

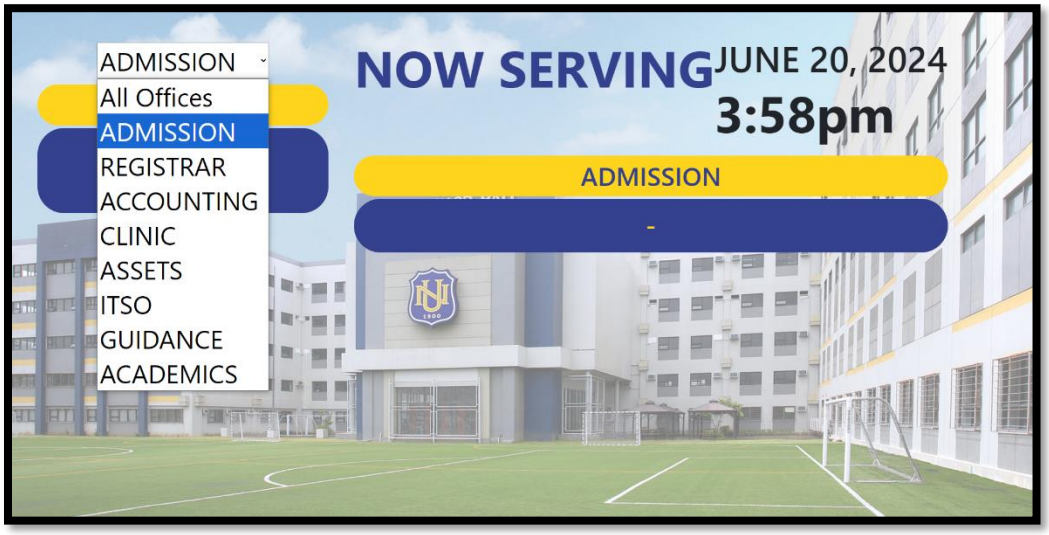
The main interface consists of the queue numbers per each office. On the left side, it has the queue numbers that are waiting to be called on its designated office.

Features:

- View queue numbers
- View queue status

Actions:

- The administrator can select which offices will be displayed on the main interface.
- On this interface, it shows all offices available along with the queue numbers waiting to be called on designated office.



- This is where the administrator selects a specific office that will be displayed on the main interface.

2.1.3 API, Libraries, and Frameworks Used

- **Bootstrap**

- **Version:** 5.3.2
- **Description:** Bootstrap is a popular CSS framework used for developing responsive and mobile-first websites. It includes CSS- and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components.
- **CDN Link:** <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-T3c6CoIi6uLrA9TneNEoa7RxnatzjcDSCmG1MXxSR1GAsXEV/Dwwykc2MPK8M2HN" crossorigin="anonymous">
- **JavaScript Bundle:** <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js" integrity="sha384-C6RzsynM9kWDrMNeT87bh95OGNyZPhcTNXj1NW7RuBCsyN/o0jlpcV8Qyq46cDfL" crossorigin="anonymous"></script>

- **Bootstrap Icons**
  - **Version:** 1.5.0
  - **Description:** Bootstrap Icons is a library of free, high-quality, open-source icon fonts. It is designed to work seamlessly with Bootstrap, but it can be used independently.
  - **CDN Link:** <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.5.0/font/bootstrap-icons.css">

- **Font Awesome**
  - **Version**: 4.7.0
  - **Description:** Font Awesome is a widely used icon set and toolkit that allows you to easily add scalable vector icons and social logos to your website.
  - **CDN Link:** <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css">

- **jQuery**
  - **Version:** 3.6.0
  - **Description:** jQuery is a fast, small, and feature-rich JavaScript library. It makes things like HTML document traversal and manipulation, event handling, and animation much simpler with an easy-to-use API that works across a multitude of browsers.

- **CDN Link:** <script src="https://code.jquery.com/jquery-3.6.0.min.js"> </script>

- **Chart.js**
  - **Version:** Latest
  - **Description:** Chart.js is a simple yet flexible JavaScript charting library for designers & developers. It supports various types of charts including line, bar, radar, doughnut, pie, polar area, and bubble.
  - **CDN Link:** <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

2.1.4 Authentication & Security

- **Single-factor authentication:** The user is required to input the correct credentials (email and password) to access the super admin module. Otherwise, they remain on the login page.

2.1.5 Common Errors

- **Data Integrity** - Exporting data in PDF and XLS formats can sometimes lead to formatting issues or data loss if not handled correctly.
- **Adding a new office –** When adding a new office, the office name should not consist of whitespaces, along with dash (-), and underscore (_) as it conflicts with the database. If adding a new office, it should format like "newOfficeName".
- **Responsiveness –** The interfaces are not totally responsive on desktop and laptop, particularly on mobile devices. When using the SUPER ADMIN module, the browser should be at maximized.

2.1.6 Recommendations

- Allow the super admin to create additional users with access to the super admin module to prevent future inconveniences caused by having only one super admin account.
- Enhance the add office function to allow spaces and apostrophes in the name or acronym of the office for greater flexibility and accuracy in office naming.
- Increase the security of the system by including two-factor authentication to prevent unauthorized access.
- To enhance system security, store passwords in the database as hashed values using a strong hashing algorithm.
- Add a change password feature to ensure the safety of the account.
- Add a forgot password feature to allow users regain access to their account.
- Improve the date filter for the reports by making it more specific using an date type input.

2.1.7 User Credentials

Displayed on the table below are the default user credentials for the existing offices in the queuing system:

Office	Username	Password
Super Admin	SuperAdmin@nu-laguna	12345678
Admission	Admission1	12345678
Registrar	Registrar1	12345678
Accounting	Accounting1	12345678
Clinic	Clinic1	12345678
Assets	Assets1	12345678
ITSO	ITSO1	12345678
Guidance	Guidance1	12345678
Academics		
ABM	ABM1	12345678
HUMSS	HUMSS1	12345678
STEM	STEM1	12345678

2.2 ADMISSIONS

2.2.1 Overview

The admissions office handles various aspects of the enrollment process as well as any other transactions involving students, freshmen, transferees, and other individuals with concerns or queries. The goal of this queueing system for the admission office is not only to provide comfortable queueing, but also to implement a well-organized and efficient queueing policy for the admission office with the help of the queueing features that this system has.

2.2.2 Functionalities & Screenshots

- **Login interface**
    - The Login feature of the system ensures the security access of users. This feature allows the admin user to access the system using their unique credentials.

Features:

- Username input field
- Password input field

Actions:

- If the admin user entered the correct credentials, it will redirect and access the admin office dashboard.

- If the admin user entered the incorrect credentials, it will prompt an error message where the user inputs the incorrect credentials.

- **Admission Office Dashboard**

The admin dashboard allows the user to view the existing queue numbers, it allows the user to endorse queue numbers to other office, End the queue number once the transaction is done, and notify the queue number to inform or notify the student who has the queue number.

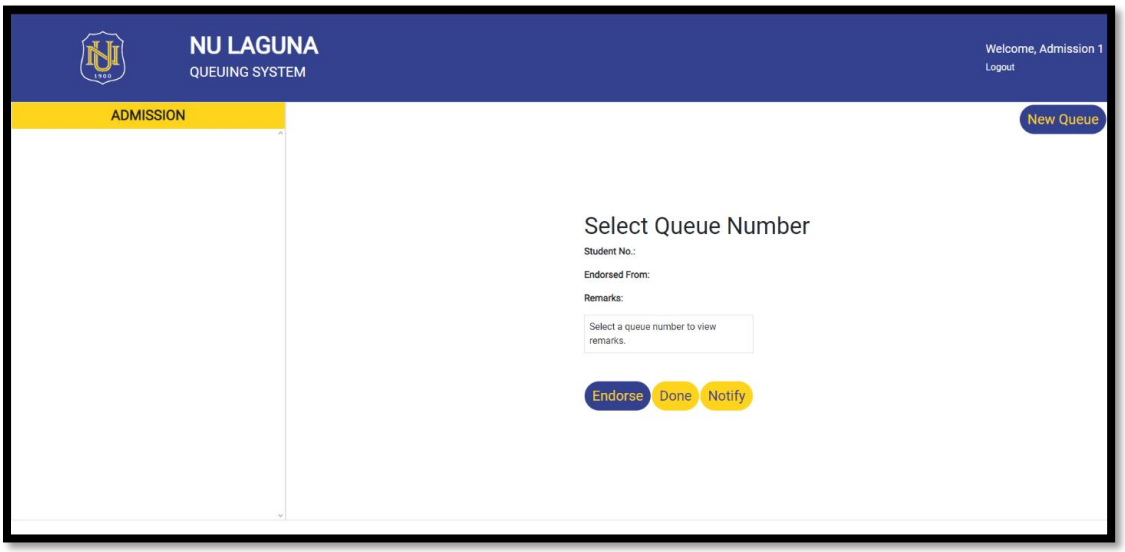
Features:

- QUEUE List
- Endorse
- End Transaction
- Notify
- New Queue

Actions:

- If the admin user selects a queue number in queue list, it will display the details of queue number in the dashboard.
- If the admin user endorsed the selected queue number, the queue number will be endorsed to the other offices.
- If the admin user ends the transaction of the selected queue number, the queue number will be gone on the queue list.
- Before doing the transactions in the dashboard after selecting a queue number, the admin user must notify the queue number first to do the transactions on the admin dashboard.

- If the admin user needs a new queue number to generate, the new queue button will redirect the user to the creation of queue number page.



- **Queue List**

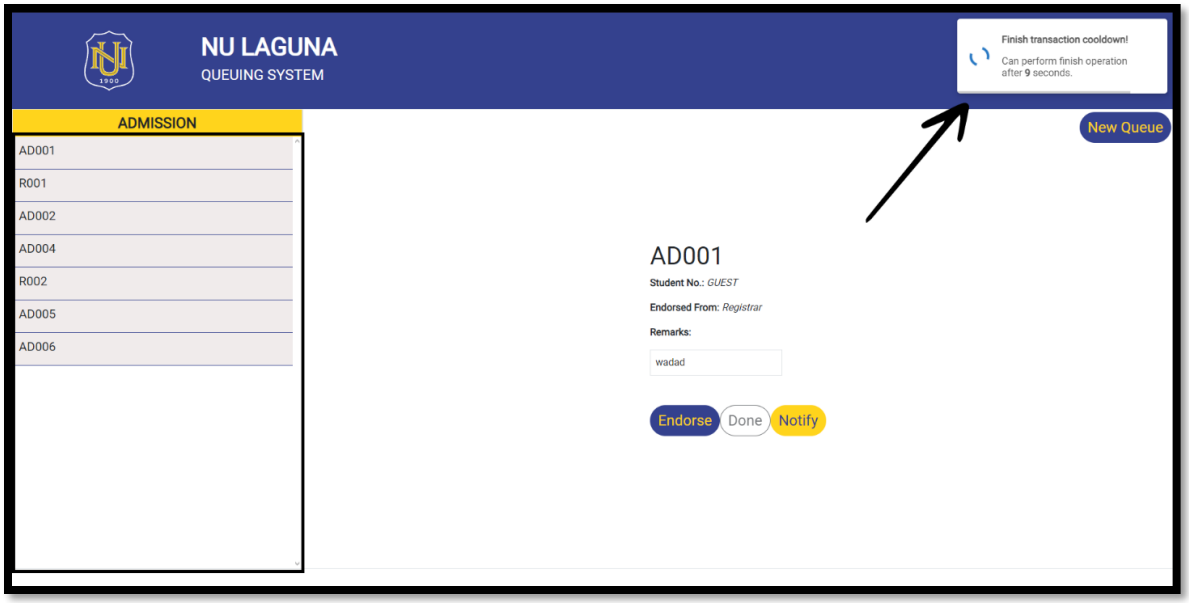
The queue list feature displays the pending lists of queue numbers. It also notifies the user that there is a ten second duration before ending the transaction of the selected queue number.

Features:

- List of queue numbers
- Notification before ending a transaction.

Actions:

- If the admin user selects a queue number in the queue list, it will display the details of the queue number in the dashboard.
- After the user admin selects a queue number, a notification will pop up that there is ten seconds duration before ending a transaction.



- **Endorsement**

The endorsement feature of the system allows the user to endorse the queue number to other offices depends on the transaction that the queue number has.

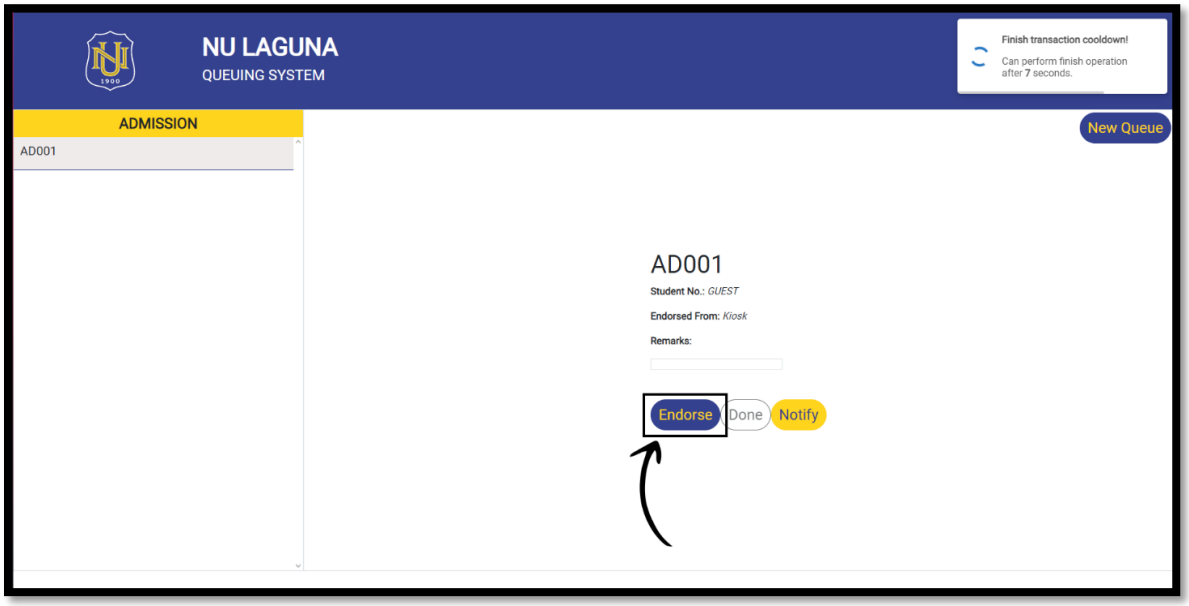
Features:

- Endorsement form
- Transaction input field
- Endorsement input field
- Remarks input field
- Confirmation of endorsed queue number

Actions:

- If the admin user clicks the endorsement button, it will display the endorsement form where it includes the transaction, endorse to, and remarks.

- If the user admin accomplished inputting the details on the required fields, it will display an alert where it says successfully endorse and the queue number will be endorsed to the other offices.



- **Completing transaction**

The Completing of transaction is a feature of the queueing system where it ends the transaction of the queue number. It displays an error validation when the user didn't input some remarks before ending the transaction of the queue number. Also, it will ask the user before ending the transaction of the queue number to avoid mistakes on this kind of matter.

Features:

- Completion of transaction form
- Confirmation alert

Actions:

- If the admin user needs to end the transaction of the queue number, a completion of transaction form will display. It will also display an error message when the admin user doesn't input the details on the required field.

- A confirmation will pop up before ending the transaction of the queue number.

- **Notify**

The notify feature of the queueing system allows the user to notify the holders of the queue number to be notified if they are on queue. It also displays the current queued number on the display queue to easily identify and notified the holder of the queue number.

Features:

- Notify

Actions:

- Allows the user to notify holders of the queue number when they are on queue.

Current Queue Display:

- Displays the current queued number on the display queue.
- Helps holders of the queue number easily identify their position and know when they are being notified.

- **New Queue**

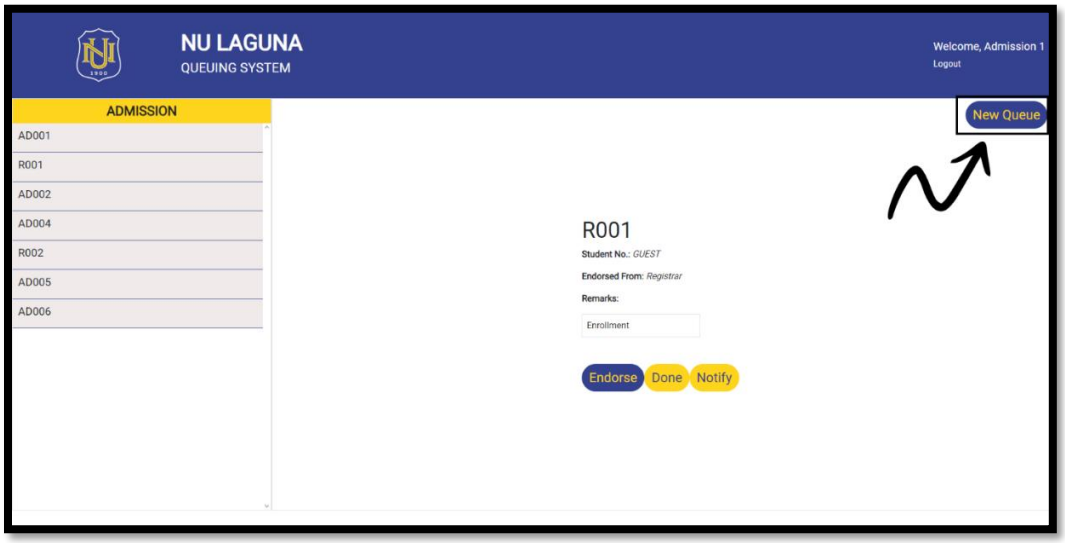
This feature allows the user to create a new queue number in case the students need a new queue number.

Features:

- New Queue

Actions:

- Allows the admin user to create a new queue number.



2.2.3 API, Libraries, and Frameworks Used

This queueing system in the Admissions office operates using session-based interactions via AJAX requests. The following libraries are utilized:

- **jQuery** (via CDN):
  - **Version**: Latest version from CDN.
  - **Description**: A fast, small, and feature-rich JavaScript library. It makes things like HTML document traversal and manipulation, event handling, and animation much simpler with an easy-to-use API that works across a multitude of browsers.
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- **Bootstrap** (via CDN):
  - **Version**: Latest version from CDN.
  - **Description**: The most popular CSS framework for developing responsive and mobile-first websites. It includes CSS- and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components.

2.2.4 Authentication & Security

- **Password Encryption**: Passwords are encrypted using the MD5 algorithm for hashing. This is a one-way hashing function that ensures passwords are stored in an encrypted format.
- **Authentication**: To access the admissions webpage, users must provide valid credentials. The system checks these credentials against the stored, hashed passwords to verify user identity.
- **Session Management**: Sessions are managed to keep track of authenticated users. If a session variable is missing, users are required to re-authenticate by logging in again.

2.2.5 Common Errors

- **Session Variables Not Present**:
  - **Description**: This error occurs when the session variables are not properly set or have expired.
  - **Resolution**: Users can resolve this issue by simply logging in again at the admissions login URL. This will re-establish the session and restore the necessary variables.

2.2.6 Recommendations

- The items on the left side (queue numbers) can be dragged and dropped to manually sort them by priority.
- UI design improvements

2.3 REGISTRAR

2.3.1 Overview

The Registrar Office Queueing System is intended to streamline and manage student service processes effectively. It starts with a secure Login Interface, where registrar administrators enter their username and password. Once logged in, the Main Interface provides a comprehensive view: the Top Bar welcomes administrators and allows them to log out; the Sidebar displays active queue numbers, which can be clicked to reveal details in the Center Panel. Notifying waiting students, endorsing transactions for other offices, and completing transactions with detailed remarks are all examples of actions. The system includes an Endorsement Form Interface for easy office-to-office transactions and an End Transaction Interface for quick service completion. Key features ensure data security and smooth operations, with APIs providing robust functionality and authentication protocols ensuring secure access.

Future enhancements may include creating a user-friendly mobile or web portal for student queuing, automated notifications, categorized queue numbers for service specificity, and a feedback mechanism to continuously improve system performance and user experience. These enhancements aim to improve efficiency, data security, and user satisfaction throughout registrar office operations.

2.3.2 Functionalities & Screenshots

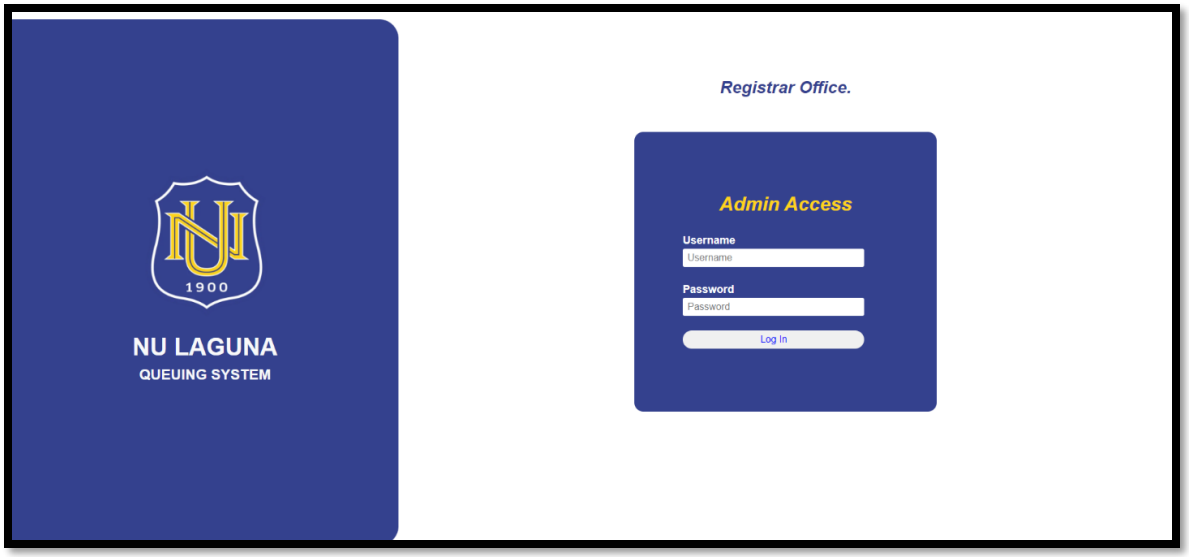
- **Login Interface**
    - The login interface provides access to the Registrar Queuing System. The registrar admin must enter their credentials to gain access to the system.

Features:

- Username Field: Input Field for the Registrar admin's username.
- Password Field: Secure input field for the Registrar admin's password.
- Login Button: Submits the entered credentials for authentication.

Actions:

- If the username and password are correct, the system will redirect to the main interface.
- If the password is incorrect, the system displays a pop-up message indicating a Login failed. Please check your username, password, or office.



- **Main Interface Registrar Office Queuing System**

  Top Bar:

  - Greetings and User Information: Displays "Hi there, [username]"
  - Logout Link: Allows users to securely log out of the system.

  Sidebar:

  - Queue Number List: Display the queue numbers that are currently active, whether launched from a kiosk or endorsed by other offices to the registrar.
  - Clickable Queue Numbers: Users can click on any queue number to view detailed information in the center panel.

- **Center Panel**

  Queue Number Details:

  - **Selected Queue Numbers:** Displays the queue number that was clicked in the sidebar.
  - **Queued Time Received:** Displays the time at which the queue number was initially received.
  - **Student Number or Guest:** Indicates the student number or marks the visitor as GUEST if no student number is available.
  - **Endorsed From:** Specifies whether the queue number was endorsed by another office or came from the kiosk.
  - **Transaction Details:** Provide an Information about the type of transaction associated with the queue number.

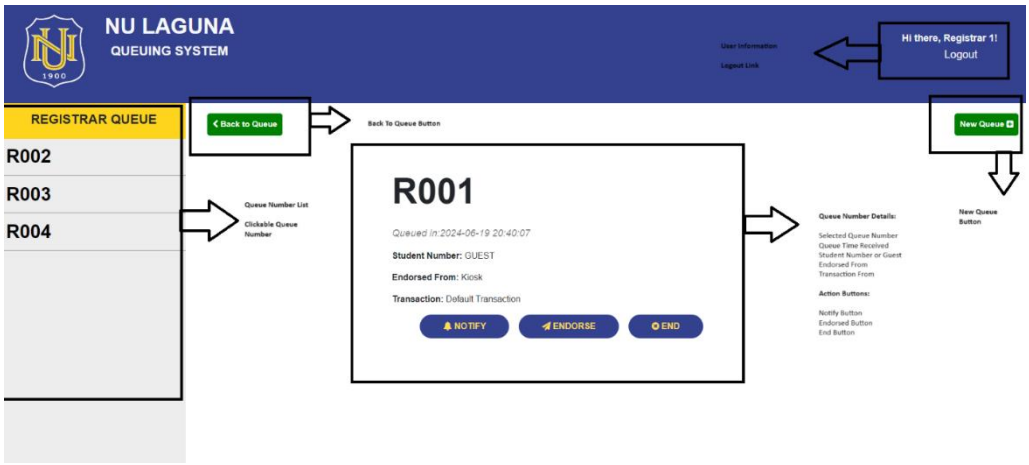
Action Buttons:

- **Notify Button:** The queue number was displayed on the TV for students and guests who were waiting.

- **Endorsed Button:** Opens the Endorsed Form to enter the necessary information for endorsing the transaction to another office.
- **End Button:** To complete and finalize the current transaction, open the End Transaction Form.
- **Back to Queue Button:** Returns the selected queue number to the sidebar, allowing other admins to access it.

Additional Features:

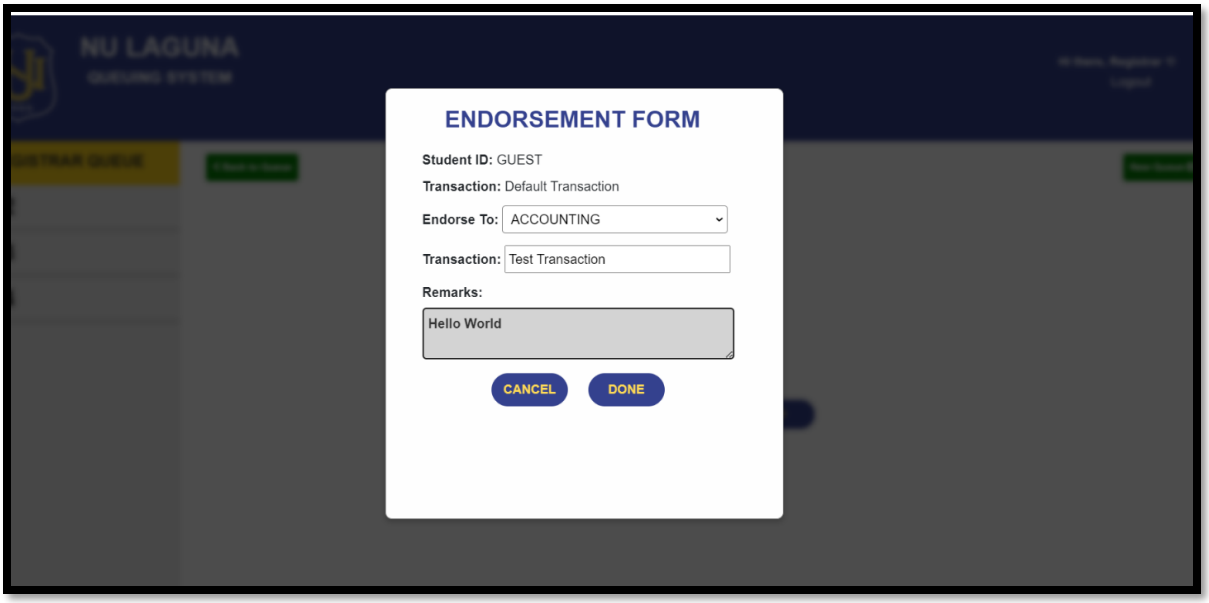
- **Launch New Queue Button:** Use the Kiosk Interface to create and launch a new queue number in the registrar's office.



- **Endorsement Form Interface**
    - **Student ID:** Depending on the selected queue number.
    - **Transaction Input:** Users can specify the details of the transaction they are endorsing.
    - **Endorsement Destination:** Enables users to select the department or office to which the transaction is being endorsed.
    - **Remarks:** Users can provide additional comments or instructions related to the transaction.
    - **Cancel Button:** Allows users to cancel the endorsement process and return to the main interface.
    - **Done Button:** Becomes clickable only when all required fields (transaction details, endorsed to, remarks) are filled out.

Detailed Actions:

- **Selected Queue Numbe**r
    - Users can view details of a transaction by clicking on its queue number from the main interface.
- **Initiate Endorsement**
    - After selecting a queue number, users click the "Endorse" button to access the endorsement form interface.
- **Fill Endorsement Form**
    - Depending on the queue number.
    - Specify the transaction details (type and purpose).
    - Choose the department or office where the transaction is being endorsed.
    - Enter any necessary remarks.
- **Validation**
    - The "Done" button is inactive until all required fields (transaction details, endorsement to, remarks) are completed.
- **Cancel Endorsement**
    - Users can click the "Cancel" button to abort the endorsement process and return to the main interface.



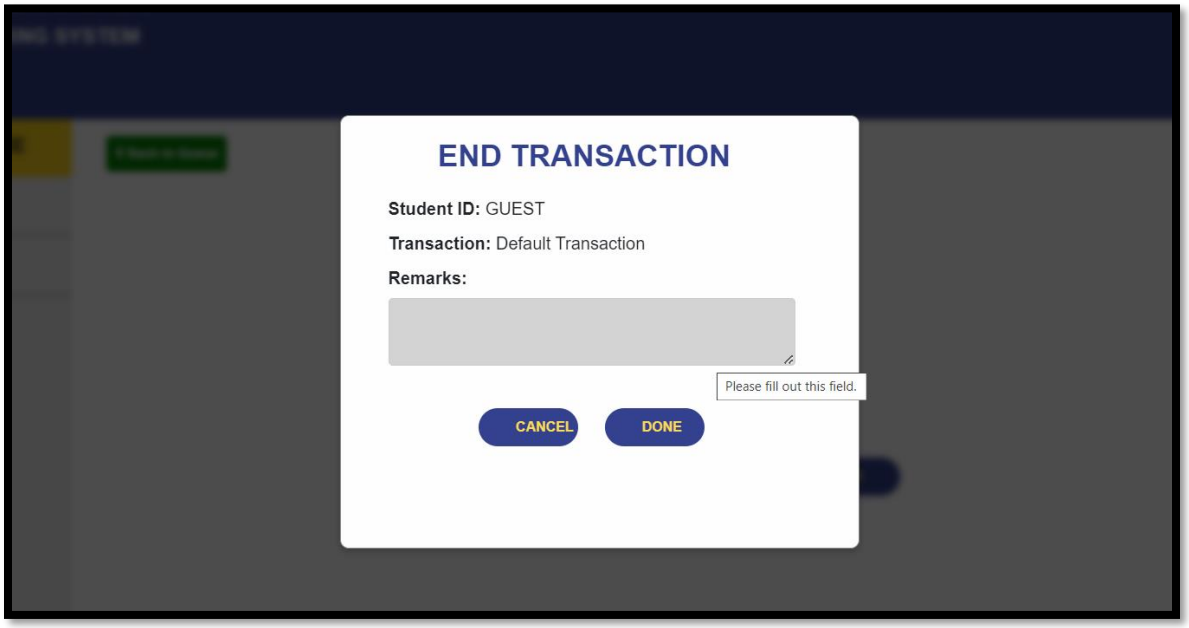
- **End Transaction Interface**
    - The End Transaction Form is accessed when a staff member clicks the "Notify" button for the next student in the queue. After 5 seconds, the "End Transaction" button becomes active, leading to this form.

Key Features:

- **Student Id:** The ID of the selected student from the queue is automatically populated.
- **Transaction Type:** Specifies the type of transaction being completed.
- **Remarks:** Field to enter notes or comments related to the transaction.
- **Cancel Button:** Allows cancellation of the transaction process.
- **Done Button:** Becomes active only after filling up the remarks field, completing the transaction process.

Functionality Overview:

- **Notify Button Behavior:** Clicking the "Notify" button initiates a countdown (5 seconds) before the "End Transaction" button becomes clickable.
- **Transaction Completion:** Users must enter remarks before they can click the "Done" button, ensuring proper documentation of each transaction.
- **Cancellation Option:** Provides flexibility to cancel a transaction if needed, returning the student to the queue.



2.3.3 API, Libraries, and Frameworks Used

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  - **Description**: A fast, small, and feature-rich JavaScript library. It makes things like HTML document traversal and manipulation, event handling, and animation much simpler with an easy-to-use API that works across a multitude of browsers.
- **Bootstrap** (via CDN):
  - **Version**: Latest version from CDN.
  - **Description**: The most popular CSS framework for developing responsive and mobile-first websites. It includes CSS- and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components.
- **AJAX** (via CDN):
  - **Version**: Latest version from CDN.
  - **Description**: A technique used in web development to create fast and dynamic web pages. It allows data to be exchanged with a web server asynchronously, meaning the page does not need to be reloaded each time data is fetched or submitted. AJAX enables developers to update parts of a webpage without refreshing the entire page, providing a smoother and more interactive user experience.

2.3.4 Authentication & Security

To ensure the authentication and security of your Registrar Office Queueing System technical document, we must address several critical aspects:

- **Authentication Mechanism**

1. **Username and Password Security**

    - Username Field - Accepts the registrar admin's username.
    - Password Field - Securely accepts the administrator's password.
    - Login Button - Initiates the authentication process.

2. **Authentication Flow**

    - When you click the Login Button, the system checks the username and password entered against the stored credentials.
    - If both are correct, the system will redirect to the main interface.
    - If the login information is incorrect, a pop-up message will appear indicating that the login failed. It should indicate whether the username, password, or both are incorrect to aid in troubleshooting.

2.3.5 Common Errors

- **Incomplete Instructions for Login**

    - Specify any requirements or restrictions related to usernames and passwords.

There are currently no errors; all functionalities are functioning properly.

2.3.6 Recommendations

- Creating a mobile application or web portal where students and parents can queue on their own other than the kiosk machine to lessen waiting time or queueing line. Including automated Notification for the user (students and parents) to be reminded when their turn is approaching.

- Creating a Categorized queue number to enhance efficiency for identifying user's needs (e.g., separate queue number for registration, inquiries, document drop-off or pick-up).

- Enhance data privacy and security by adding secure login and data protection for administrators.

- Creating a feedback loop regarding the registrar, queuing system, mobile application/ web portal (if created) to further enhance and to identify problems and produce solutions regarding said problems and issues

2.4 ACCOUNTING

2.4.1 Overview

The Queueing system for Accounting Office is intended to make it more convenient for students in making a payment transaction or other concerns or queries regarding the payment. The Accounting office would verify the student's receipt if they made a payment and after the verification, Accounting have option to endorse the student to other offices. This system improves efficiency by reducing waiting time and to streamline the process within the accounting office.

2.4.2 Functionalities & Screenshots

- **Login**
    - Accounting staffs are required to Login their account that ensures the security access of users. This feature allows accounting staffs user to access the system using their unique credentials.

Features:

- Username input field
- Password input field
- Login in button

Actions:

- If the accounting staff entered the correct credentials, it will redirect and access the admin office dashboard.

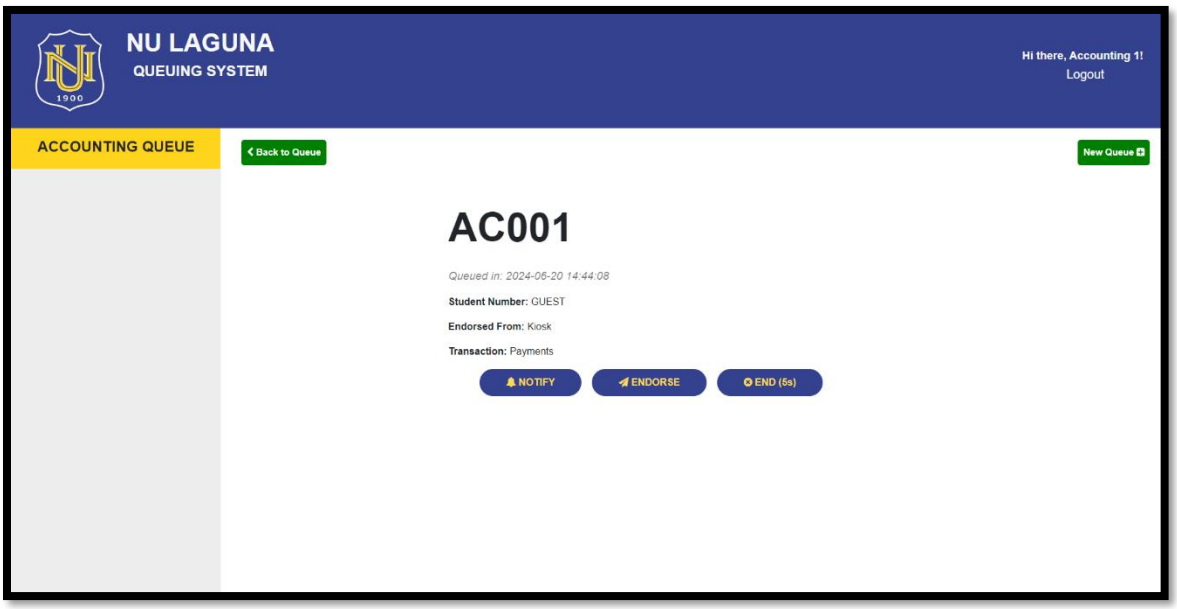
- If the accounting staff entered the incorrect credentials, it will prompt an error message where the user inputs the incorrect credentials.

- **Accounting Office Dashboard**

Accounting office dashboard allows accounting staffs to view the existing queue numbers, allows the user to endorse queue numbers to other office, End the queue number once the transaction is done, and notify the queue number to inform or notify the student who has the queue number.

Features:

- QUEUE List
- Endorse
- End Transaction
- Notify
- New Queue
- Back to queue



Actions:

- If the user selects a queue number in the queue list, it will display the details of queue number in the dashboard.
- If the user endorsed the selected queue number, the queue number will be endorsed to the other offices and would disappear in the queue list.

- If the user ends the transaction of the selected queue number, the queue number will be removed in the queue list.
- Before doing the transactions in the dashboard after selecting a queue number, the admin user must notify the queue number first to do the transactions on the admin dashboard.
- If the user needs a new queue number to generate, the new queue button will redirect the user to the creation of queue number page.
- If in an instance that the user had clicked the wrong queue number, clicking the Back to queue button will return the queue number back to the queue list.

- **Queue List**

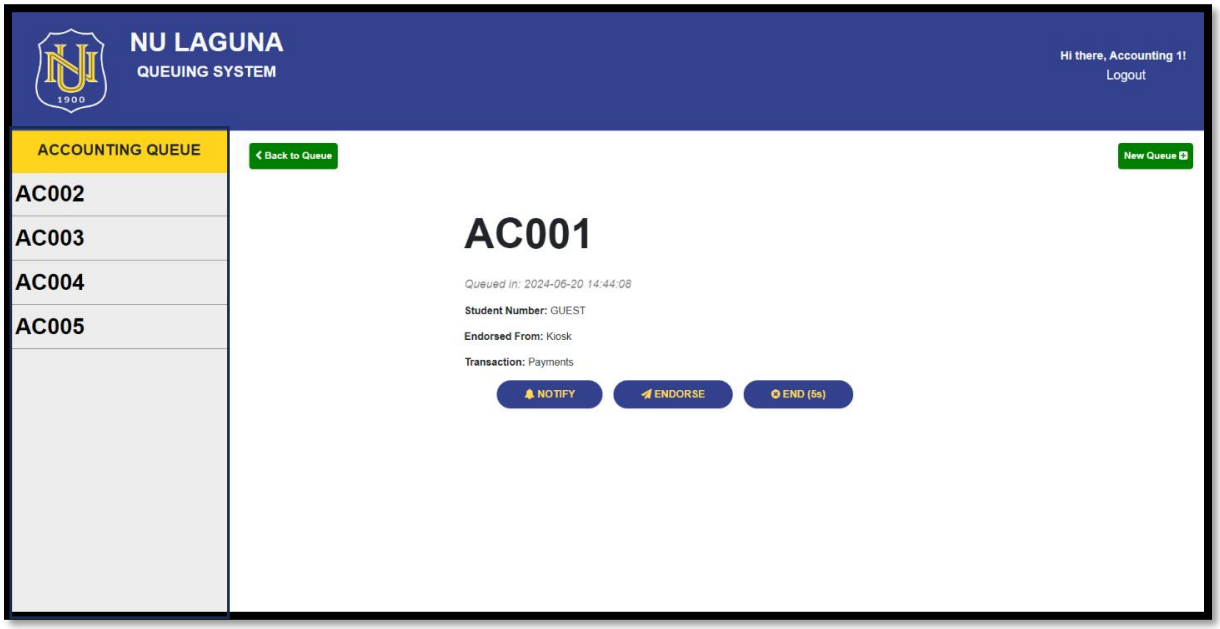
The queue list displays all the pending lists of queue numbers coming from the kiosk or endorsement from other offices.

Features:

- List of queue numbers
- Notification before ending a transaction.

Actions:

- If the user selects a queue number in the queue list, it will display the details of the queue number and would be removed in the queue list and transferred in the dashboard.



- **Endorsement**

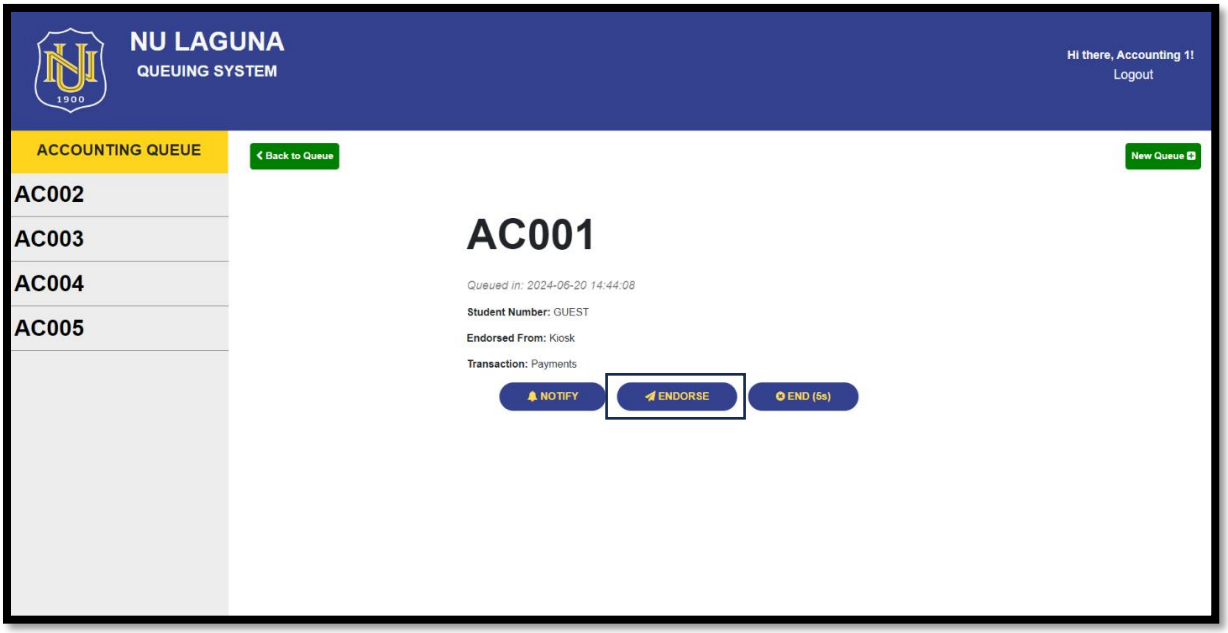
The endorsement feature of the system allows the users to endorse and pass the queue number to other offices depending on the transaction that the queue number and student needs.

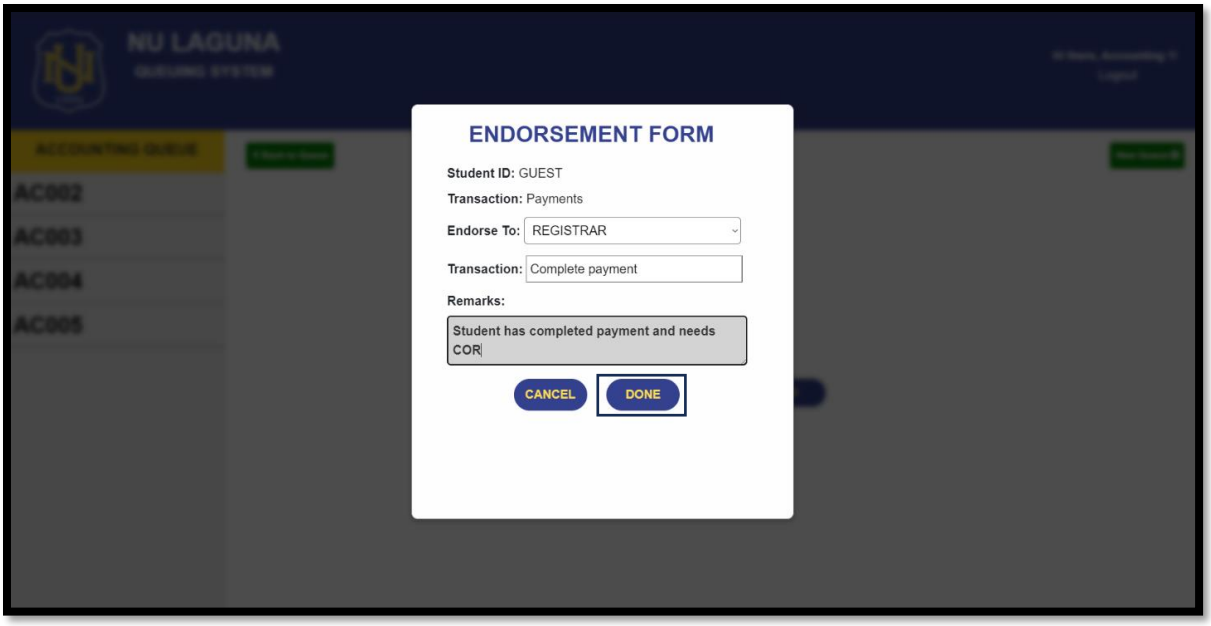
Features:

- Endorsement form
    - Transaction input field
    - Endorsement input field
    - Remarks input field
- Confirmation of endorsed queue number

Actions:

- The users need to notify the queue number first to display the current queue number in the display queue monitor signifying that the number is being called before endorsing to other offices.
- If the user clicks the endorsement button, it will display the endorsement form where it includes the transaction, endorse to, and remarks.
- Once done filling up the form and ready to submit, clicking done will pass the queue number to the selected office and the current number will be removed in the dashboard.



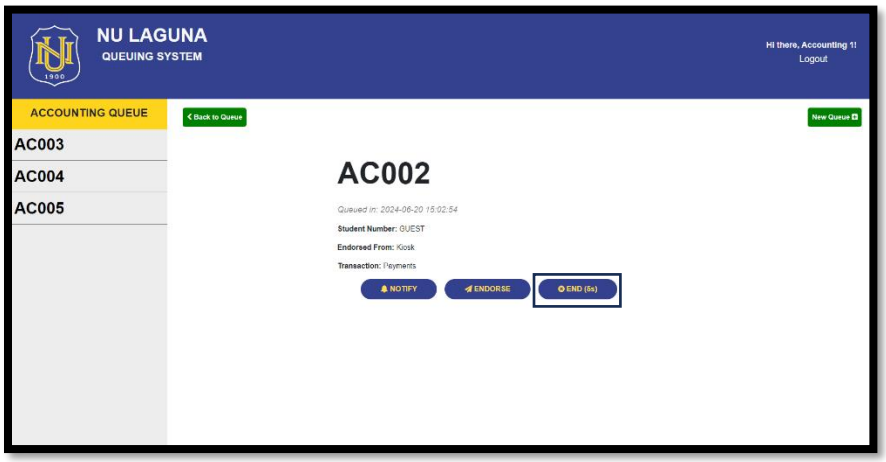


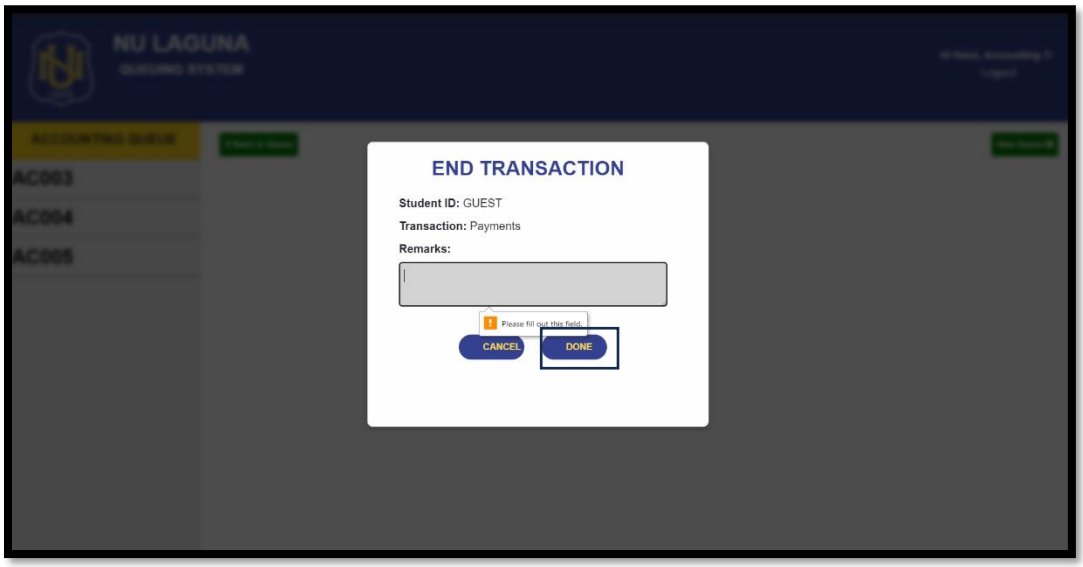
- **End Transaction**

The end transaction or End button allows users to end a transaction once it is done and no need to endorse to other offices.

Actions:

- A cool down of 5 seconds renders the button unclickable, this prevents any mishaps resulting in any accidental ending of a queue number.
- Upon clicking the end button an end transaction form would be displayed, remarks are required to be filled up before proceeding to end the transaction.
- Once the user has filled up the remarks, clicking the done button would result in removal of the queue number as it is considered done.



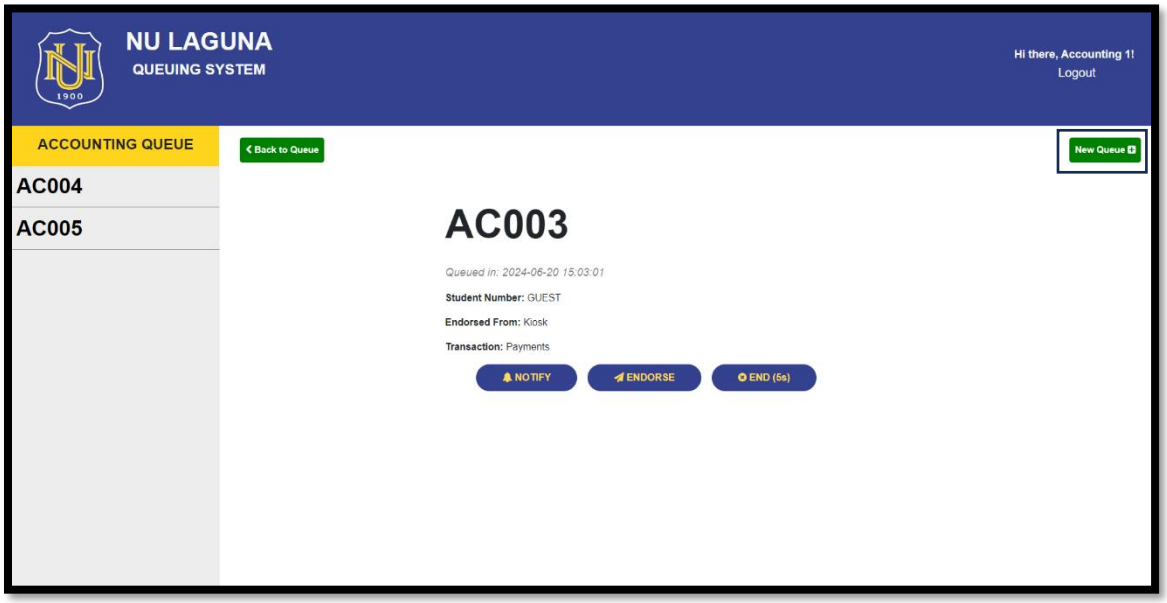


- **New queue**

New queue allows users to manually add a queue using their device in an instance where there is a need of manual queuing of students.

Actions:

- Clicking the new queue button will redirect the user to the kiosk dashboard where the user can manually input the queue of the student.



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2.4.4 Authentication & Security

- **Authentication**: To access the accounting webpage, users must provide valid credentials. The system checks these credentials against the stored, hashed passwords to verify user identity.

2.4.5 Common Errors

- Invalid URL Link

2.4.6 Recommendations

- User interface improvements

2.5 OTHER OFFICES

2.5.1 Overview

The queuing system is designed to manage and streamline the flow of students through various essential service areas within the school's "Other Offices" (Assets, Guidance, ITSO, and Clinic). The system ensures that each office operates efficiently and that individuals receive timely assistance. It manages queues for medical assessments and consultations in the Clinic, handles the distribution and requisition of school uniforms and other merchandise in the Assets office, addresses technological needs as well as updating student IDs, taking ID photos, and resolving school account issues in the ITSO, and provides counseling services addressing psychological, academic, and social needs in the Guidance office.

2.5.2 Functionalities & Screenshots

- **Login Interface**

  The login interface provides access to the Other Offices Queuing System. Every office admin must enter their credentials to gain access to the system. Other Offices use Guidance as the Main Interface.

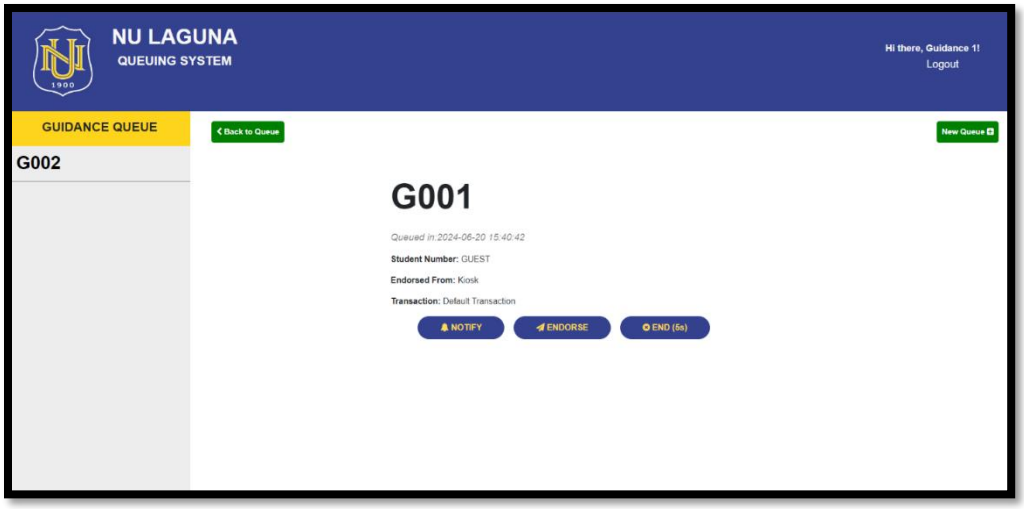
  Features:

  - **Username Field**: Input Field for every office admin's username.
  - **Password Field**: Secure input field for every office admin's password.
  - **Login Button**: Submits the entered credentials for authentication.

- **Admin Home Interface**

Features:

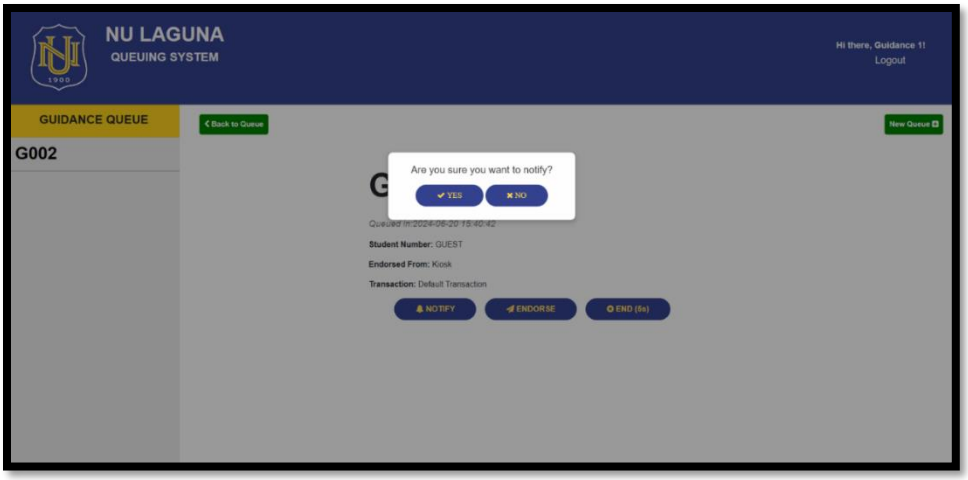
- **Back to Queue -** It allows the admin to return the selected queue from the list.
- **End -** This function automatically ends the queue's transaction.
- **Endorse -** This function allows a queue number to be endorsed to different offices.
- **Notify - W**hen clicked, the selected queue number will be called on and displayed in the display monitor.
- **New Queue –** It allows the admin to get the new queue.
- **Queue List –** It contains the queue numbers of queries in the Guidance office.



- **Notify Interface**

Functionality:

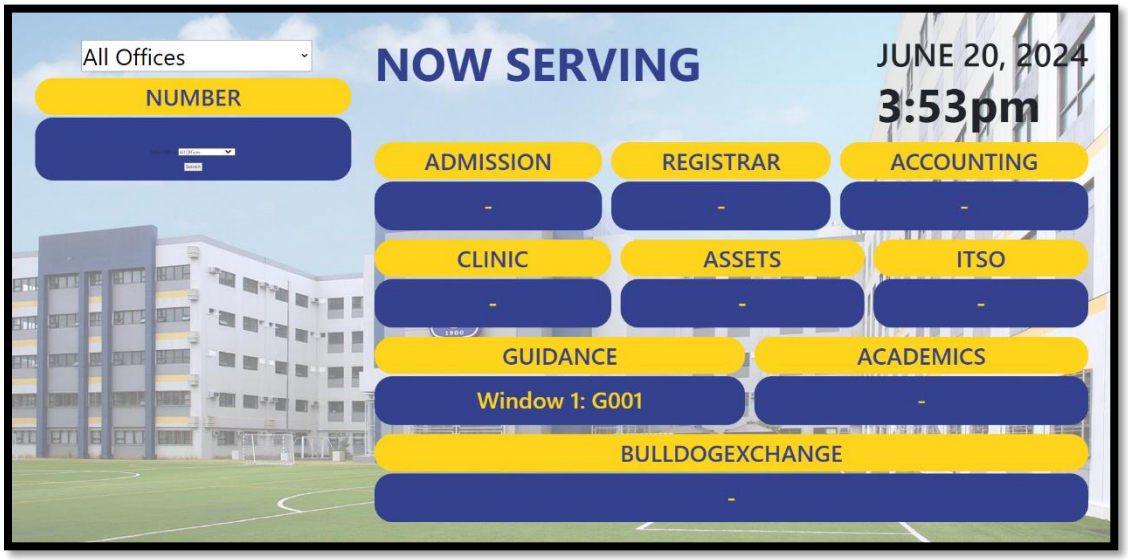
- When the Notify button is clicked, it will show the admin two options: **Yes** (to continue to Notify) or **No** (cancel the transaction).



- **Display Queue Interface**

  Features:

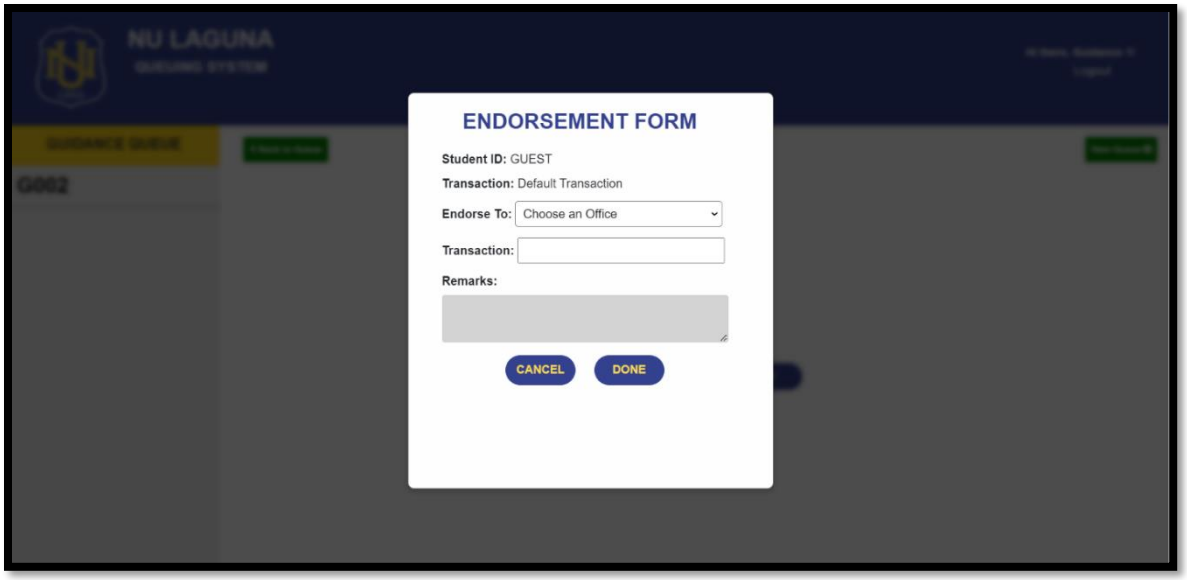
  - It shows the Queue number the offices Guidance notifies before endorsing or end transaction and the customer about their queue number process.

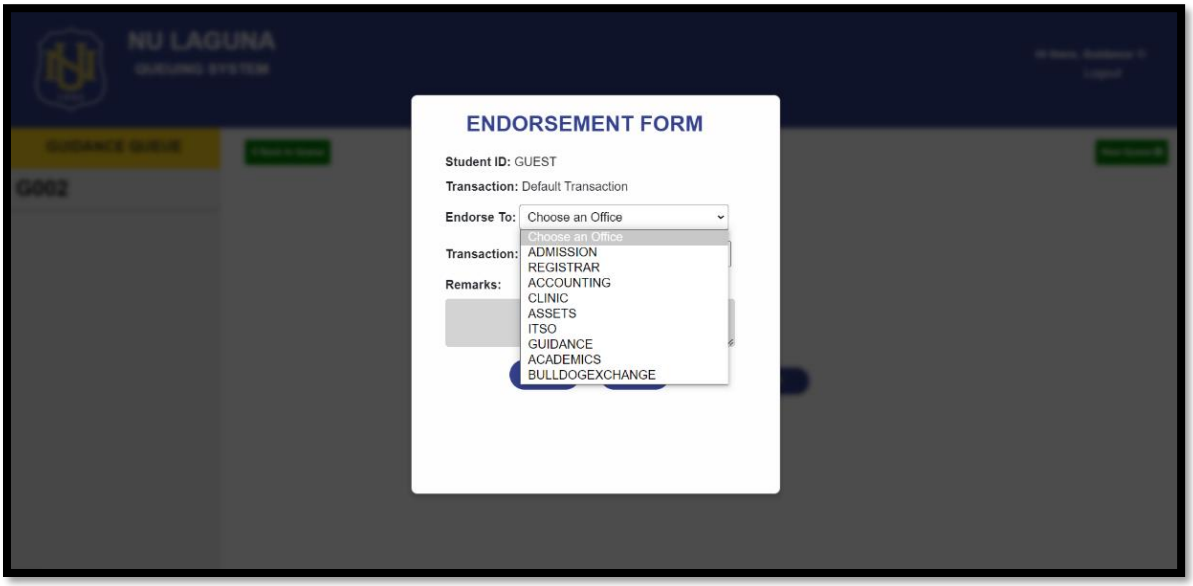


- **Endorse Transaction Interface**

  Features:

  - **Endorsement Form -** It displays a form to be filled out to endorse a different office.
  - **Endorse To -** This part is where the admin chooses the office which the queue will be endorsed to.
  - **Remarks -** allows the admin to add other remarks regarding the transaction to be endorsed.
  - **Student ID -** This displays the Student ID, if it is from a Guest, it will display Guest.
  - **Transaction -** The transaction part specifies the transaction to be made hence why it is endorsed.



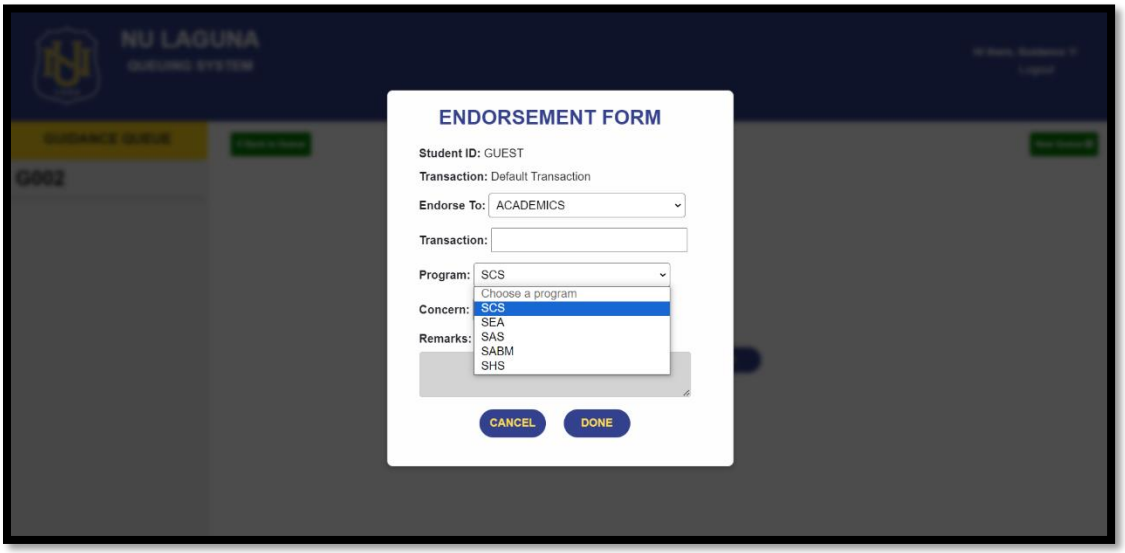


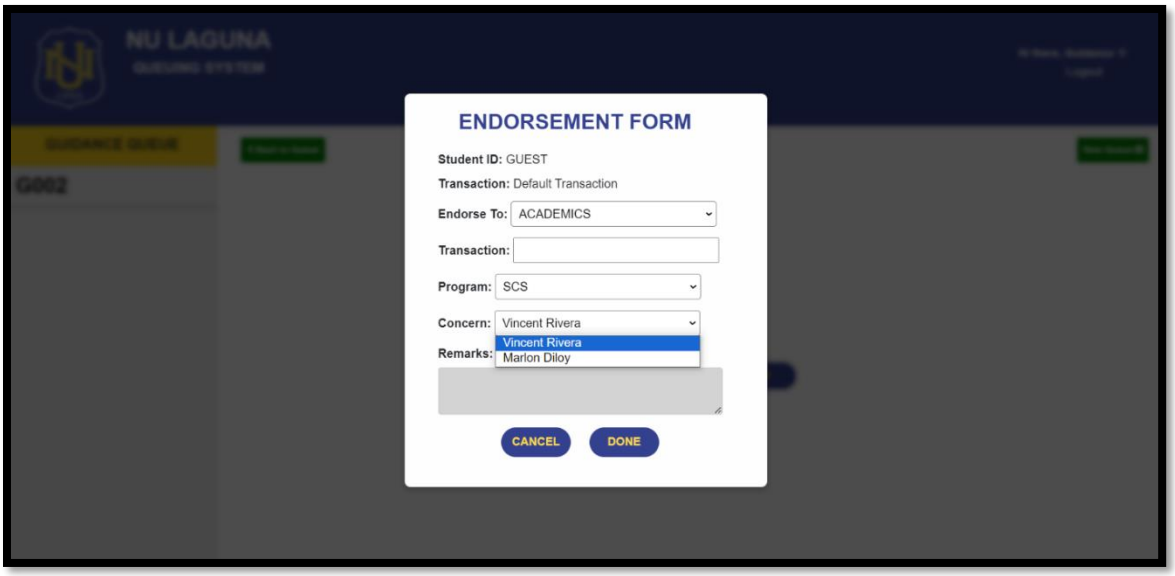
Features For Academics:

- **Concern –** It allows the admin to choose the designated Deans for the concern.
- **Program –** It allows the admin to choose the department.

- **Functionality:**
  - First you will choose the offices that you endorse and fill out the blank form for the details of transactions.
  - When endorsing Academics, it will need additional information: Program, as well as the Concern of the customer in visiting the designated Deans.

- When you click Done the Queue number will go to the office that you choose to endorse.





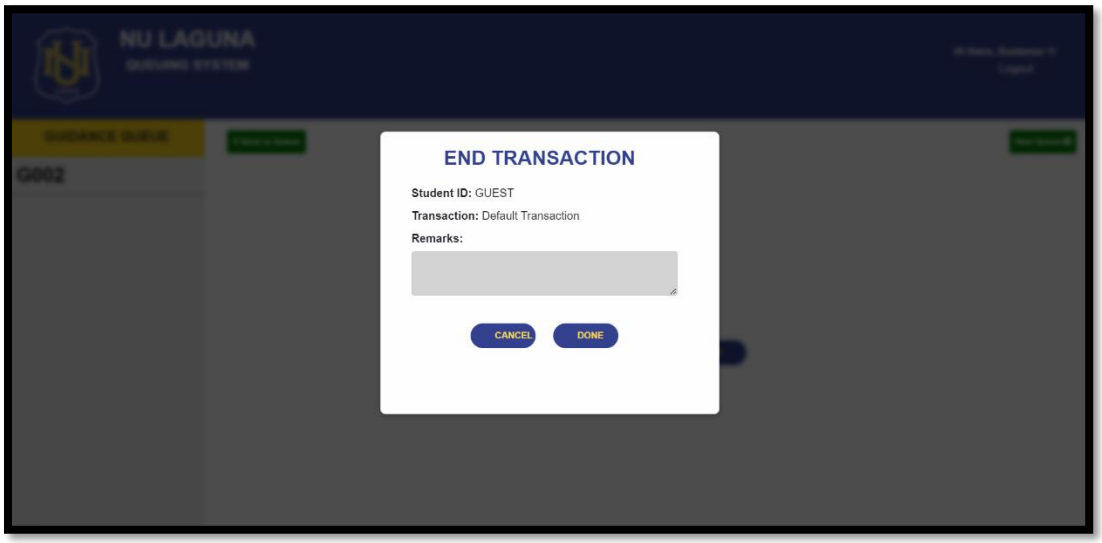
- **End Transaction Interface**

  Features:

  - **Student ID -** It allows the admin to know if the customer is an existing student or applicant.
  - **Transaction & Remarks –** It allows the admin to know what kind of transactions that happened.

Functionality:

- Admin fill out the remarks to know about the transactions or the concern is accommodated.



2.5.3 API, Libraries, and Frameworks Used

- **Bootstrap**
  - **Version:** 5.3.2
  - **Description:** A front-end framework prioritizing responsive design, ensuring uniformity and adaptability across various devices for all office interfaces.

- **jQuery**
  - **Version:** 3.7.1
  - **Description:** A JavaScript library that streamlines manipulating DOM and managing events, enabling interactive features and AJAX requests across all modules within the offices.

- **SweetAlert2**
  - **Description:** A JavaScript library that generates attractive and customizable alert dialogs, offering user-friendly notifications and confirmations across the Queuing System.

- **AJAX**

- **Description:** For rapidly generating interactive web pages, enabling asynchronous communication with the server. This facilitates updating specific parts of a webpage without reloading the entire page.

2.5.4 Authentication & Security

- **Password Security Encryption** – In this system the password has been encrypted by MD5 algorithm to secure and avoid password leaked.

- **Login Authentication** – If the Username and Password is correct, it will direct in the Offices Interface. But if the details inserted are incorrect, it requires you to insert the correct details.

2.5.5 Common Errors

- There are no errors available. All functionality is working properly, except for Incomplete Instructions for Login. Specify any requirements or restrictions related to usernames and passwords.

2.5.6 Recommendations

- **Mobile App Integration**

  - Developing a mobile app version makes it more convenient for students to queue for their chosen offices using their mobile phones at any time. This enhances accessibility and user experience.

- **Enhanced Security Measures**

  - Implement role-based access control to ensure that only authorized personnel can access specific functions and data within the system. This enhances security and maintains data integrity.

2.6 ACADEMICS

2.6.1 Overview

The Queueing system for Academics Office is intended to make it more convenient for students to get assistance and guidance from the program chair, dean, or other academic faculty. Students will choose the department and program chair depending on their concerns. They will receive a queue number and wait for their turn. For program chairs, they can filter and manage queues by department, notify the queue numbers, endorses to other offices, and end the queue number once finished.

This system improves efficiency of academic office operations by reducing waiting time, physical congestion, and optimizing the use of faculty time, thereby streamlining processes within

the academic office. By leveraging modern technology, it enhances the overall student experience and creates a more organized, efficient, and user-friendly environment for both students and academic staff.

2.6.2 Functionalities & Screenshots

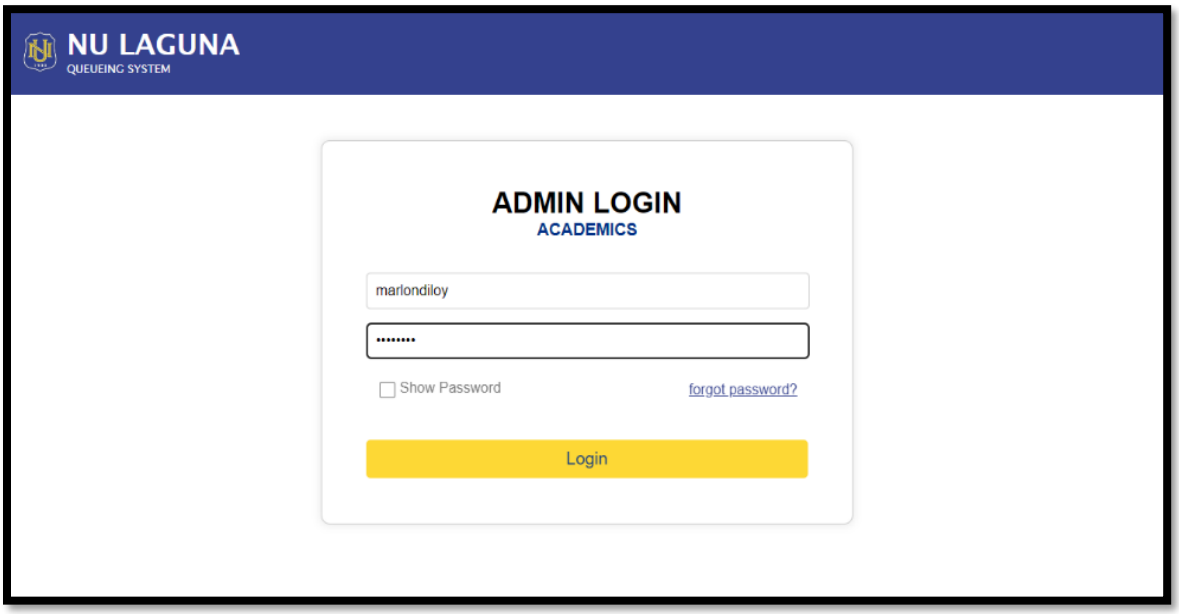
- **Log In Page**

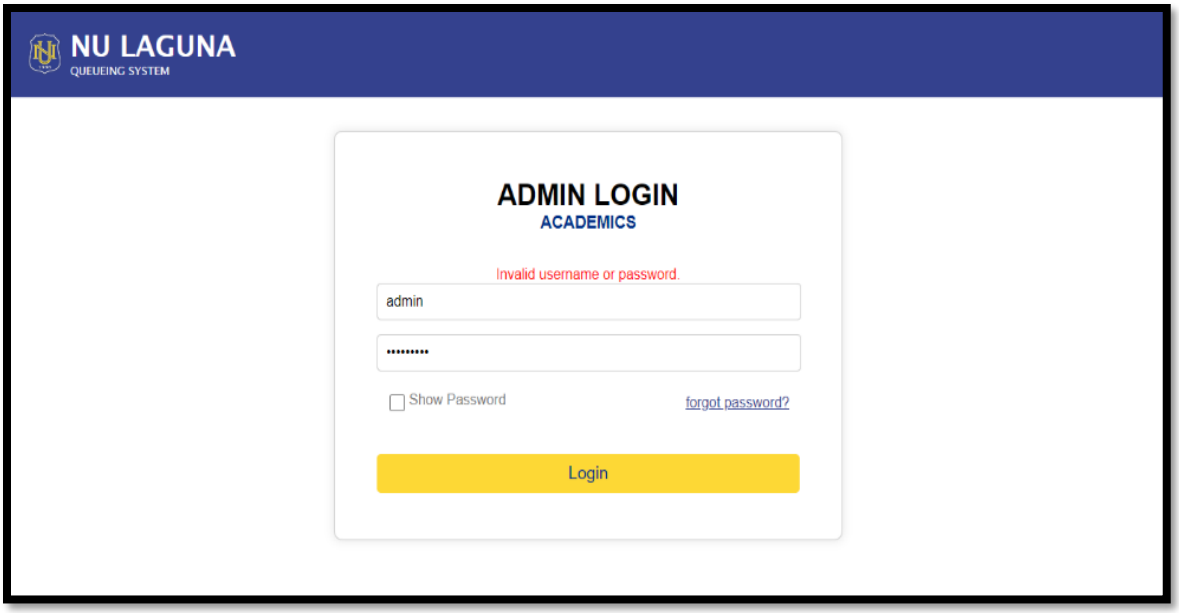
  Features:

  - Username Input Field
  - Password Input Field

  Actions:

  - If the correct credentials are entered, the user will be redirected to the academics' queuing homepage.
  - A checkbox for *"show password*" is provided for the user to double check the *hidden* characters of the password they have entered.
  - Clicking the *"forgot password?"* would redirect the user to a new page and allow them to set up a new password for their account.





- **Set-up New Password**

  Features:

  - Username Input Field
  - Password Input Field
  - Confirm Password Input Field
  - Visibility Toggle
  - Submit Button

  Actions:

  - There will be two password input fields, one for the new password and one for confirming the new password. The system will alert the user if the password does not match.
  - There is a visibility toggle where users can show and hide their passwords to avoid typographical errors.
  - Clicking the submit button will automatically change the user's password.

- **Academics Queuing Home Page**

  Features:

  - List display of active queue numbers fetched from the system's database.
  - Display of information of the selected queue numbers:

    - *Student I.D.*: Student's I.D. number, but if the user is a guest, the information will be assigned as, "GUEST."
    - *Transaction*: Selected kind of transaction from the kiosk or the office that the user was endorsed from. This should be what's the student/guest's agenda to the current office.
    - *Endorsed From*: The office where the user was before the current transaction.
    - *Remarks*: Other information written by the last personnel of the previous office will be listed here.

  - New Queue Button
  - Return to Queue Button
  - Action Buttons: Notify, Endorse, End
  - Display of notification for new tracking number

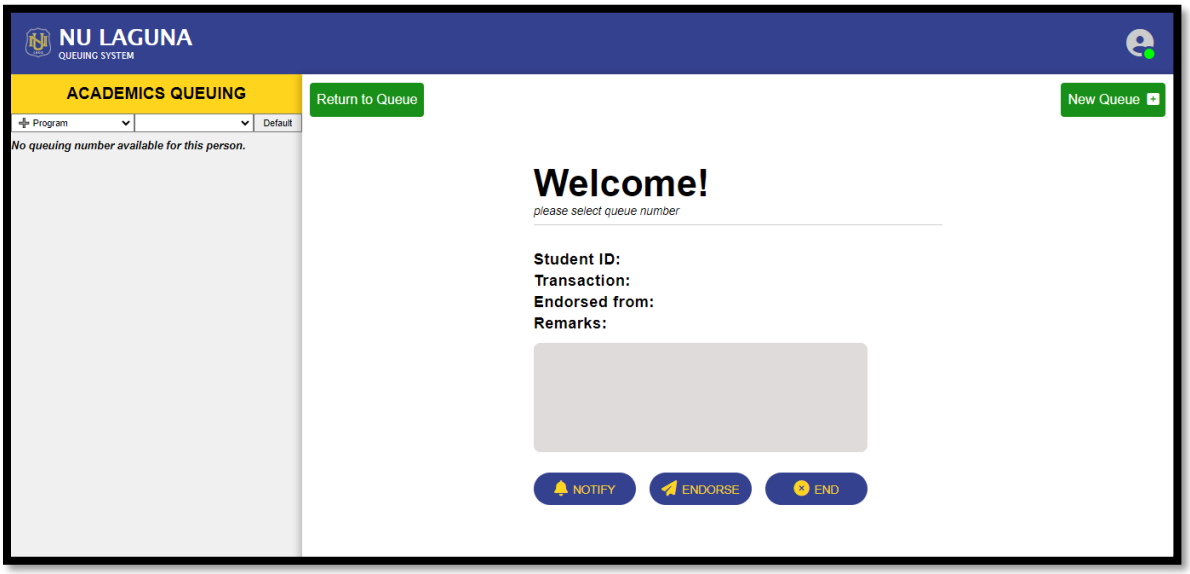
  Actions:

  - When the user selected a queue number from the academics queuing list, the system will display the assigned information to the student or guest's queuing number for the program chair to access.

- During the on-going transaction, the program chair will have to utilize the three action buttons:

    - NOTIFY – Once a queue number is selected and displayed, the program chair will have to notify it first in order for the queue number be called on to the display screen where students and/or guests may see.
    - ENDORSE – When the student or guest arrives to the office of the program chair and the transaction has been completed, they may endorse the student or guest's queuing number to the next office.
    - END – If the student or guest's transaction has no agenda with other offices, the program chair may end the transaction. Or if the student or guest have not arrived for a specific time, the program chair may also end the transaction.

- In accessing other queue numbers from a different school or program chair, the filter at the academics queue list may be utilized. The filter will have a dropdown of the schools available in NU Laguna as well as its corresponding program chairs.
- The "Return to Queue" button is an action for the displayed queue numebr to be returned to the list and be also accessible to other program chairs.
- The "New Queue" button will redirect the program chair to the Kiosk's main page where they can also select and make a new queuing number for the student or guest.



- **NOTIFY Modal**

  Features:

  - CANCEL Button
  - OK Button

  Actions:

  - When the program chair is ready for a new transaction, they may click the NOTIFY button from the homepage and a modal will pop out the screen. Selecting "OK" will automatically display the queue number in the display screen.
  - Once the queue number is displayed, a timer will start. This way, the program chair will have a grace period in waiting for the student or guest. It also helps to avoid intentionally ending the current transaction with the student or guest.
  - The END button on the homepage cannot be selected if the queue number is not notified by the program chair.
  - Selecting the "CANCEL" button will close the modal.

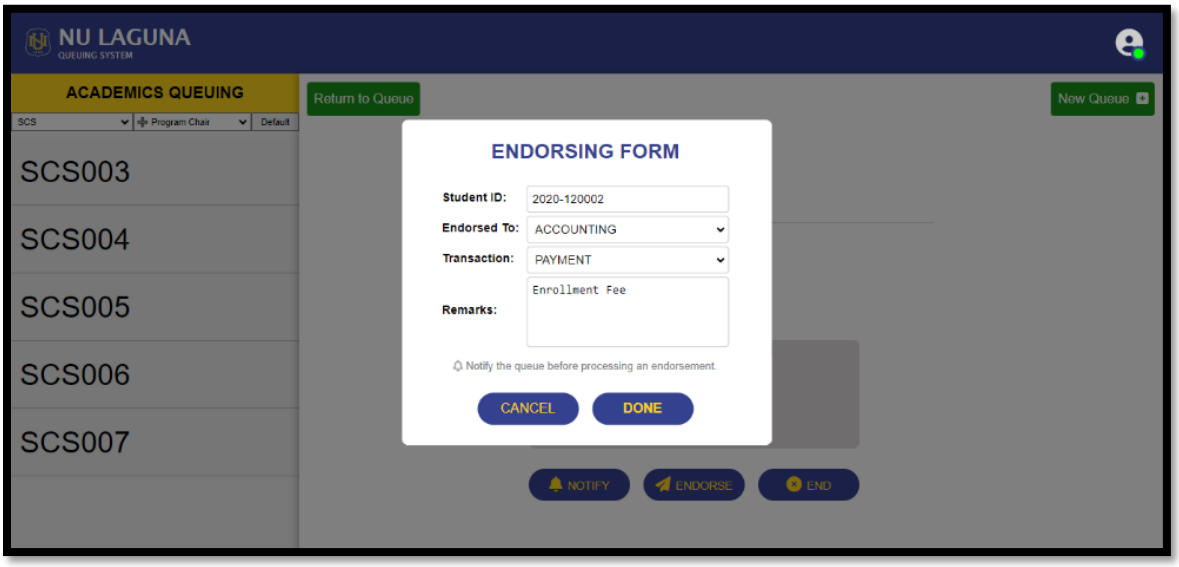
- **ENDORSE Modal**

  Features:

  - Student I.D. Display
  - Endorsed To Dropdown Menu
  - Transaction Dropdown Menu
  - Remarks Input Field
  - CANCEL Button
  - DONE Button

Actions:

- Fill up the form with the corresponding information needed for the student or guest's transaction to the next office.
- The Student I.D. may not be changed as it was encoded during the first step at the Kiosk.
- A dropdown menu is provided to only give the option of offices that the program chair usually endorses the student or guest to during the enrollment process.
- The input field for remarks are for other information that is needed for the student or guest's agenda with the next office.
- Once the form is completed, the program chair may select "DONE" and the queue number will be displayed before being sent to the next office.
- Selecting the "CANCEL" button will exit the modal.



Confirmation Modal

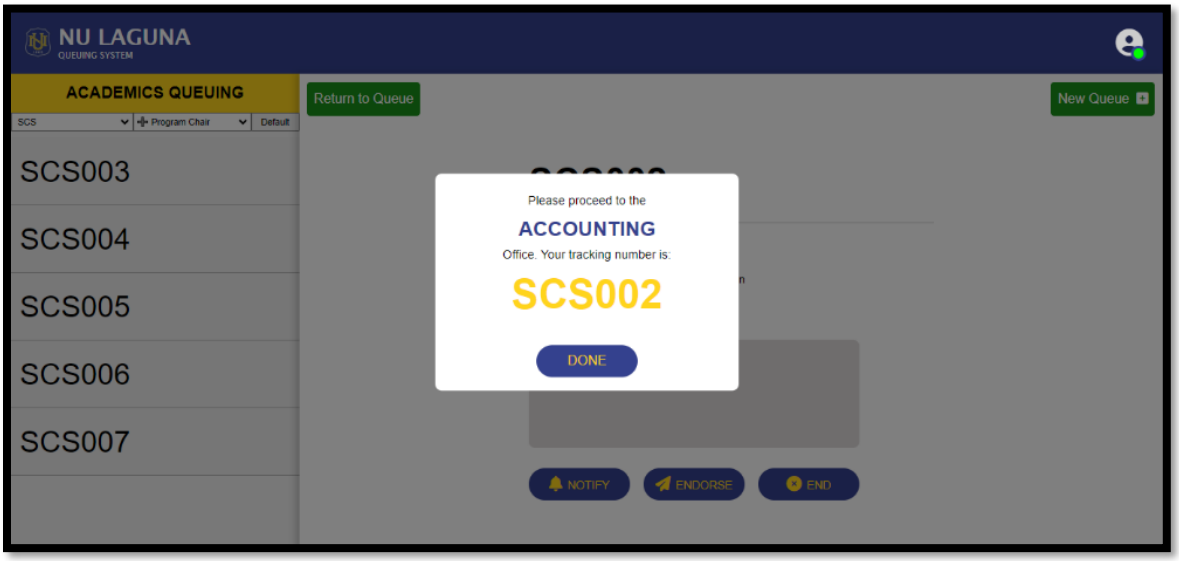
Features:

- Display of Next Endorsed Office
- Display of Current Tracking Number
- DONE Button

Actions:

- The student or guest may confirm the office to which they will proceed. The tracking number displayed will still be the same as the printed queuing number they received at the Kiosk.

- Selecting the “DONE” button will automatically endorse the queuing number to the next office and the transaction with the program chair is officially done.



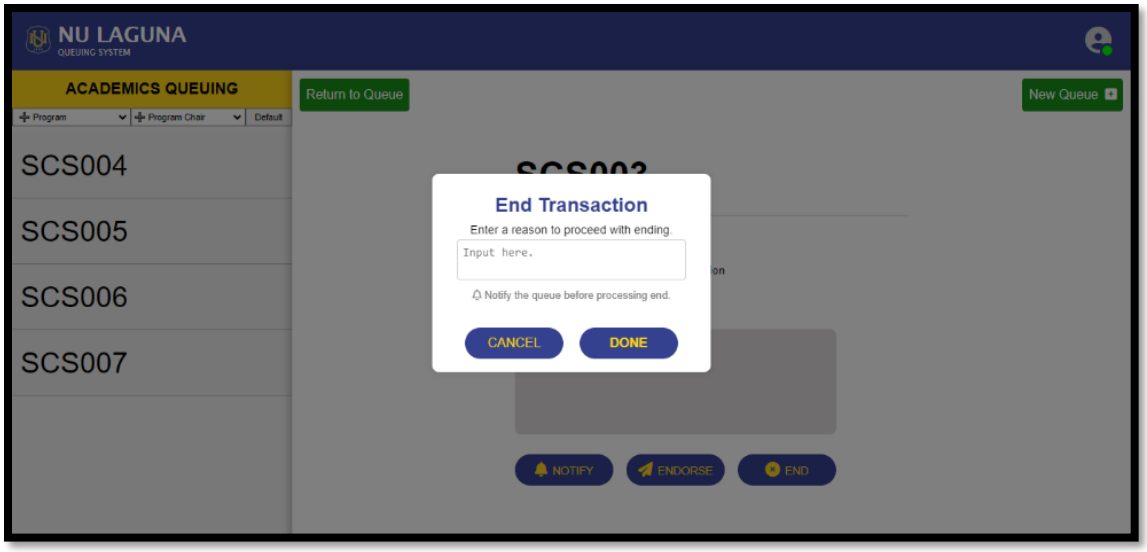
- **END Transaction Modal**

  Features:

  - Remarks Input Field
  - CANCEL Button
  - DONE Button

  Actions:

  - Before officially ending the transaction, the program chair is required to enter remarks as to why the transaction ended in the Academics Office.
  - Once sure with the encoded remarks, the program chair may select the “DONE” button to end the transaction.
  - Selecting the “CANCEL” button will exit the modal.



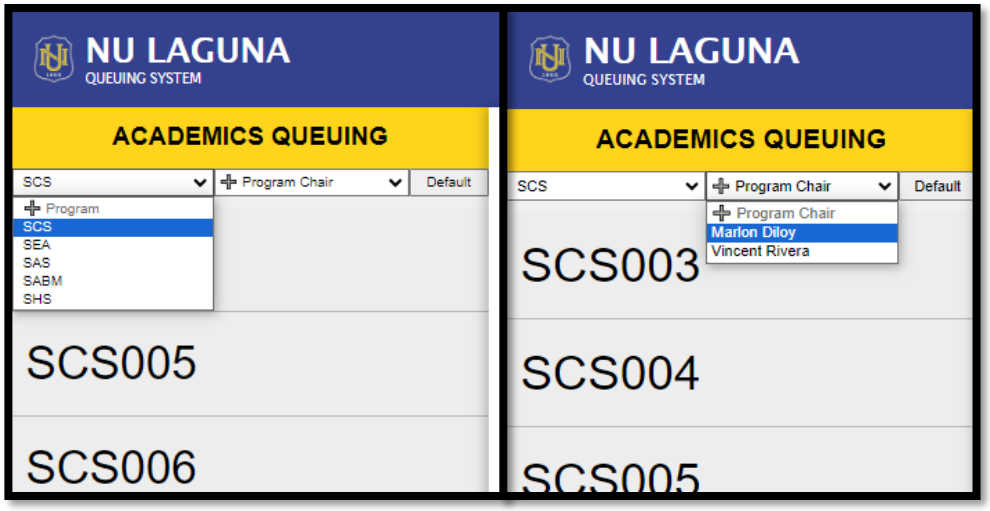
- **Academics Queue Filtering**

  Features:

  - Dropdown Menu for “Program” and “Program Chair”
  - Default Option

  Actions:

  - Program chairs can select specific a program from the dropdown menu.
  - Available options for dropdown menu:
    - School of Computer Studies (SCS)
    - School of Engineering and Architecture (SEA)
    - School of Computer Studies (SCS)
    - School of Arts and Sciences (SAS)
    - School of Accounting and Business Management (SABM)
    - Senior High School (SHS)
  - If program chair selects SCS, they might access or manage queues related to the School of Computer Studies (SCS)
  - For program chair dropdown menu, if user select SCS program, it allows them to choose between different program chair within the School of Computer Studies (SCS)



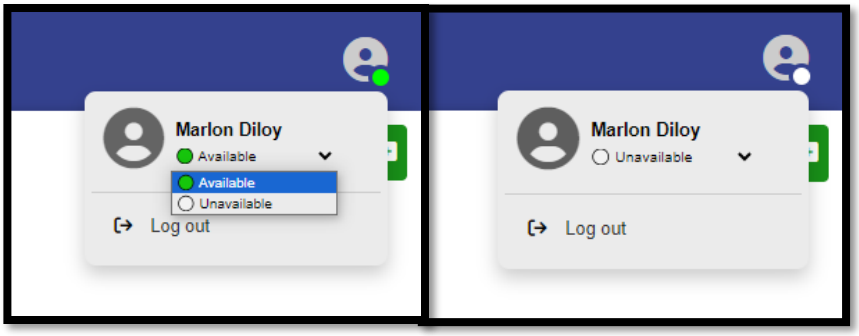
- **Program Chair Profile Display**

  Features:

  - Users Name and Icon
  - Dropdown Menu
  - Availability Status Indicator
  - Logout Option

  Actions:

  - Program Chair's Name will be displayed in the modal together with their availability status.
  - Users can change their availability status by selecting:
    - "Available" to indicate that the user in currently online or available. (With green dot indicator)
    - "Unavailable" to indicate that the user in currently offline or unavailable. (With gray dot indicator)
  - "Logout" button allows users to log out their account or session.



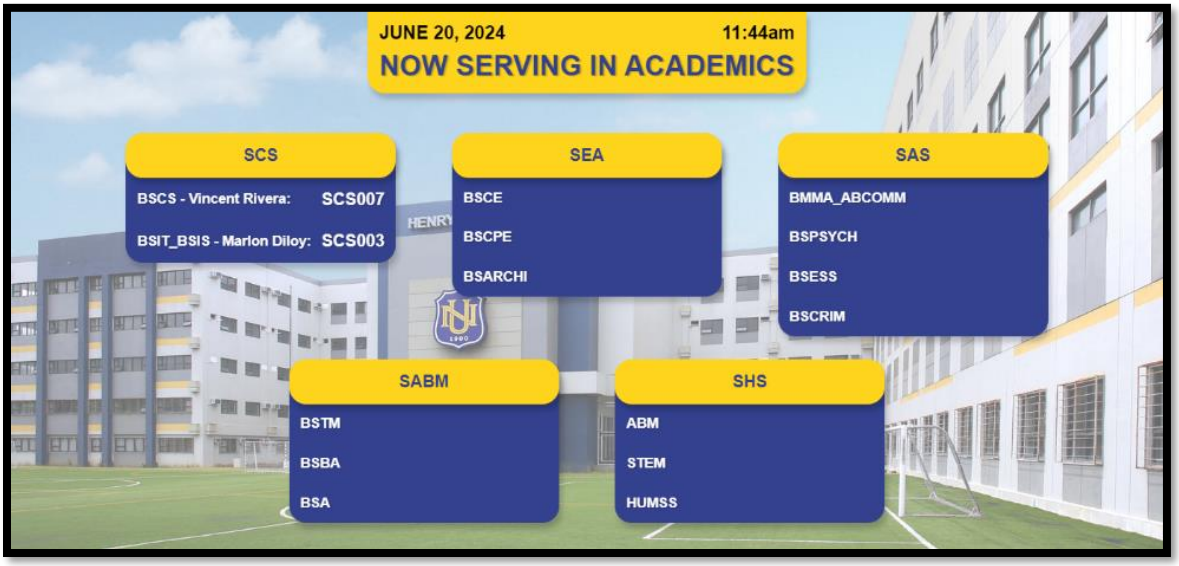
- **Display Screen for Queue Numbers**

  Features:

  - Real-time Display for Time and Date
  - Real-time Updates for Queues for Various Academic Programs
  - Alert Sound Feature

  Actions:

  - When new queue numbers are added, viewers will be notified by sound.
  - The system will only display the current queue related to its program chair.



2.6.3 API, Libraries, and Frameworks Used

In this module, the academics used AJAX to be able to send and receive data from the database asynchronously allowing it to get real-time updates and smoother user experience without requiring page reloads. Furthermore, the following are the libraries that were used in this module:

- **JQuery:**

  - **Version:** v3.7.1 via CDN (Latest version)
  - **Brief description:** a JavaScript library that is feature-rich, compact, and quick. With an easy-to-use API that functions across several browsers, it simplifies tasks like event handling, animation, Ajax, and traversing and manipulating HTML documents.

- **Document Object Model (DOM) API**:

  - **Version:** Latest Version

- **Brief description:** an XML and HTML document application programming interface (API). It outlines the logical organization of documents as well as how they may be accessed and changed. The term "document" is used broadly in the DOM standard because XML is being used more and more to represent a wide range of information types that may be stored in various systems; much of this information is often viewed as data rather than documents. However, this data is presented by XML as documents, and it may be managed using the DOM.

2.6.4 Authentication & Security

- **Authentication**: To access the academic queueing system, the user must enter their correct and valid credentials. Once the user's inputs are correct, the system will check whether the entered credentials are valid and existing inside the database or not to verify the user's identity and validity.
- **Password Encryption**: The developed system is built with a hashing algorithm known as md5. This algorithm makes the stored or created password inside the database encrypt as a hash value to avoid possible cyber security problems such as data privacy breach.

2.6.5 Common Errors

- Common Errors that were found in the module are the following:
    - Notification modal always pops up.
    - Filtering bugs.
    - Return to queue function bugs when you click another queue number.
    - The forgot password feature prompts "password resets successfully" regardless of the entered username and password and lacks a way to return to the login page if opted not to reset password.

2.6.6 Recommendations

- Forgot Password Feature:
    - The system prompts "password reset successfully" regardless of the entered username and password and lacks a way to return to the login page.
    - Recommendation: Remove the "Forgot Password" feature to prevent these issues. Alternatively, improve the feature with proper verification and security measures.

- Profile Section:
    - The profile section shows the logged-in user and a status indicator for the Dean's availability, but it is not fully functional.
    - Recommendation: Continue developing this feature as it benefits students and guests by informing them of the Dean's availability.

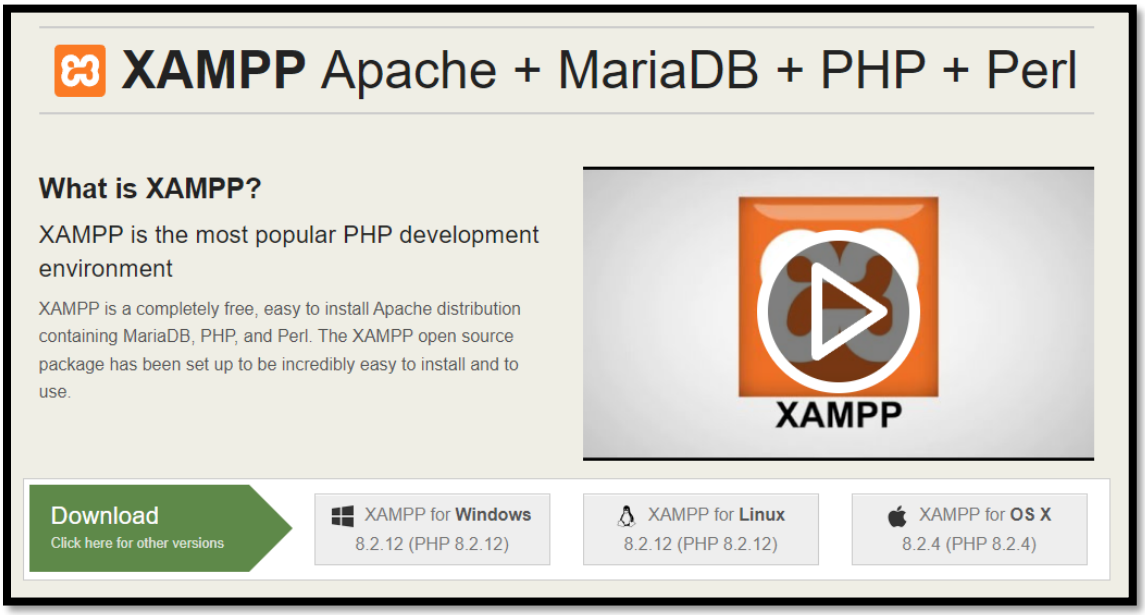
- Queue Number Selection:
    - When a queue number is clicked, other queue numbers should be non-clickable for 5 seconds, and the "return to queue" button should be disabled or removed.
    - Recommendation: Implement this to prevent multiple selections and ensure a smoother user experience.

- Pop-Up Notification:
    - The notification appears for new tracking numbers but reappears once closed.
    - Recommendation: Improve this so once closed, it doesn't reappear unless a new tracking number is entered, reducing user interruptions.

3. INSTALLATION AND SETUP

1. XAMPP Installation

- Download and install XAMPP from the official website.
- Follow the installation wizard to set up Apache (webserver), and MySQL (database).



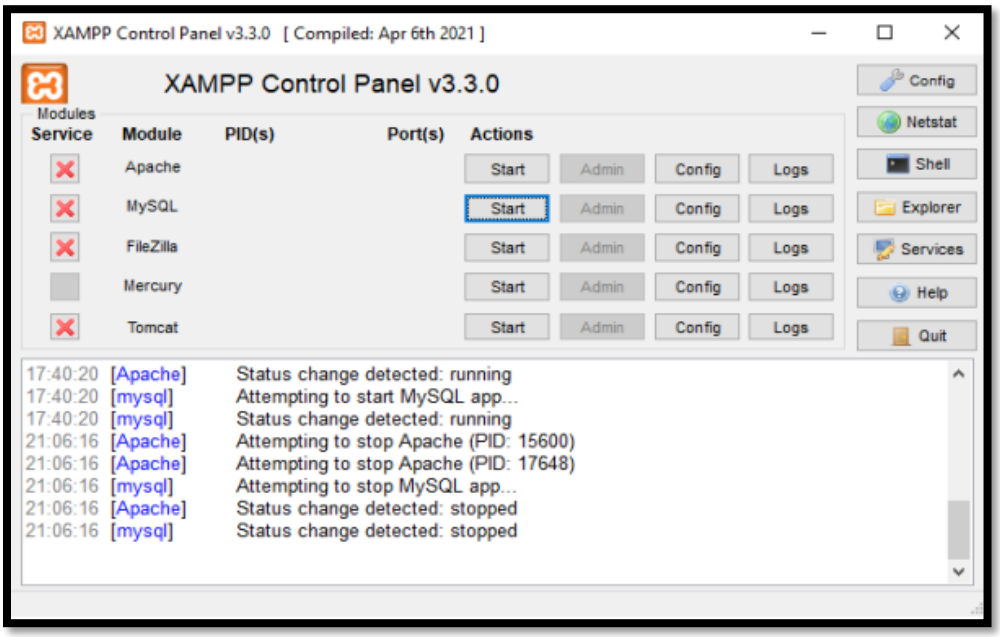
2. Queueing System Installation

- Extract the zipped Queueing System Files
- Extract the contents of the queuing system file to the XAMPP's **"htdocs"** directory.

academics	18/06/2024 10:11 pm	File folder	
accounting	18/06/2024 10:11 pm	File folder	
app	18/06/2024 10:11 pm	File folder	
asset	18/06/2024 10:11 pm	File folder	
assets	18/06/2024 10:11 pm	File folder	
clinic	18/06/2024 10:11 pm	File folder	
guidance	18/06/2024 10:11 pm	File folder	
itso	18/06/2024 10:11 pm	File folder	
node_modules	18/06/2024 10:12 pm	File folder	
otheroffices	18/06/2024 10:12 pm	File folder	
registrar	18/06/2024 10:12 pm	File folder	
sadmin	18/06/2024 10:12 pm	File folder	
script	18/06/2024 10:12 pm	File folder	
sound	18/06/2024 10:12 pm	File folder	
styles	18/06/2024 10:12 pm	File folder	
vendor	18/06/2024 10:12 pm	File folder	
.env	18/06/2024 10:11 pm	ENV File	1 KB
academics	18/06/2024 10:11 pm	Opera GX Web Do...	7 KB
academics	18/06/2024 10:11 pm	PHP Source File	5 KB
academics-display	18/06/2024 10:11 pm	Opera GX Web Do...	2 KB
academicsinterface	18/06/2024 10:11 pm	PHP Source File	9 KB

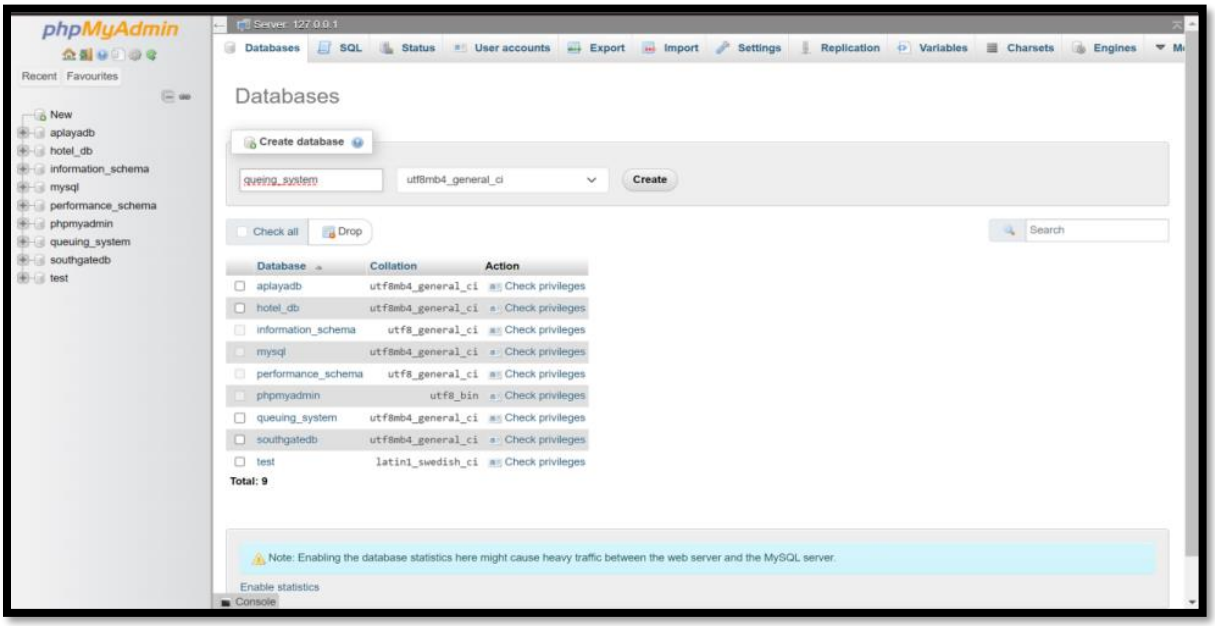
3. Start the XAMPP

- Start the Apache and MySQL services.
- Click the Admin in MySQL



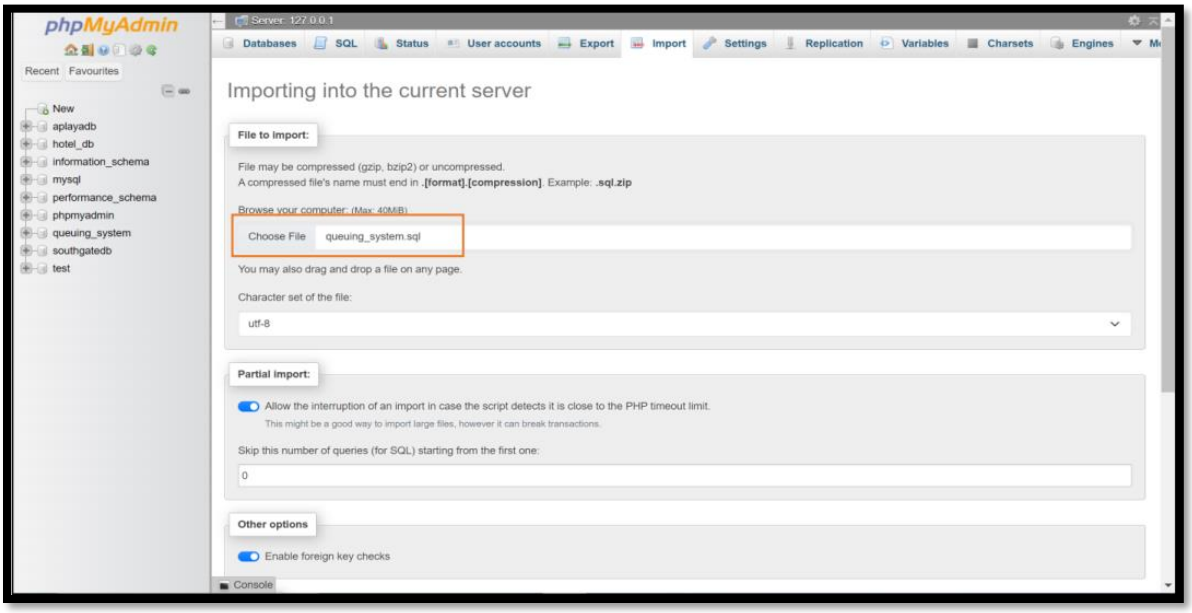
4. Database Setup

- On phpMyAdmin, create a new database for the queueing system.



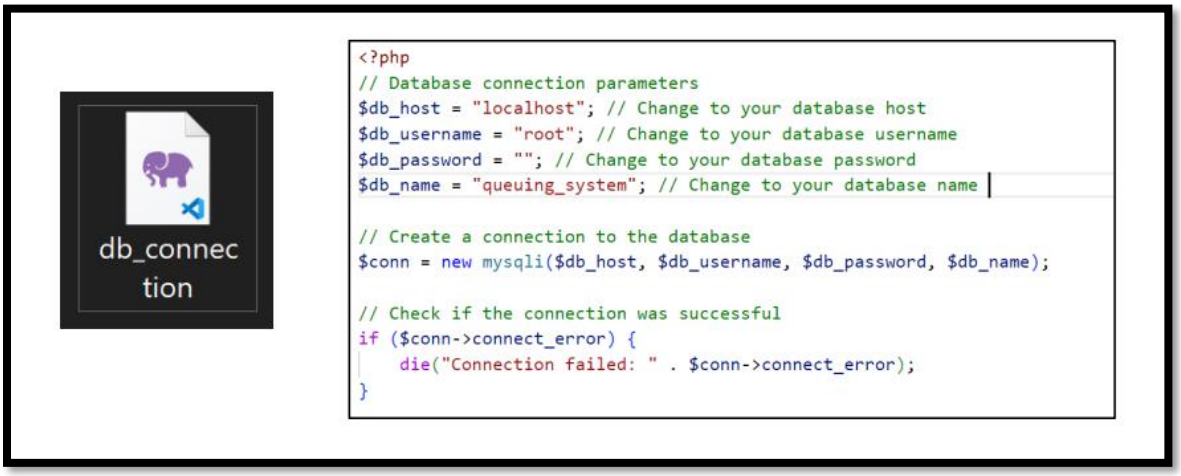
5. Import Database Schema

- Import the provided SQL file to create the necessary tables and structure in the database.



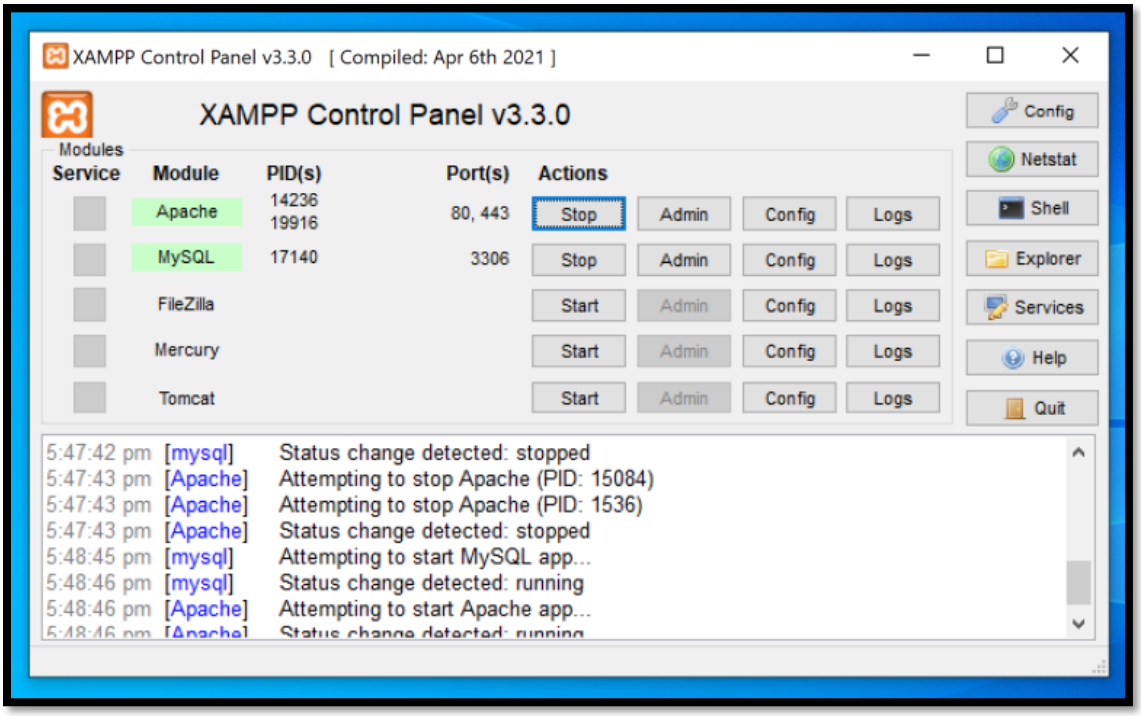
6. Configure Database Connection

- Locate the configuration file (e.g., database.php) in the queuing system files.
- Update the database connection settings with the database name, username, and password.



7. Start XAMPP Services

- Open the XAMPP Control Panel.
- Start the Apache and MySQL services.



8. Setting up Kiosks

- There will be two (2) Kiosks stationed on different parts of the campus. These kiosks will require the following:
    - Touch Screen Monitor
    - HDMI Cable
    - Lan Cable *(if applicable)*
- One will be stationed at the entrance and the other beside Inspire Lounge.
- Both Kiosks will be connected to the local host that will display the Main UI of the Queueing System. This will allow students select which office they want to be queued at.

9. Setting up the Offices

- The offices will be provided with the local host URL that will require their credentials given by the Super Administrators.
- Each office will be provided with a User Manual to better understand how the Queueing System works.
- They will also be provided with a receipt printer for those who opt to be queued in the respective office to avoid going back and forth.
- For the notification sound, the computer or machine that is assigned to project the queueing overview should first click anywhere inside the DOM as JavaScript requires interaction first before allowing delegated events like playing a sound.
- There will be five (5) TVs in total to monitor the queues. These will be installed in:
    - Sampaloc Lane

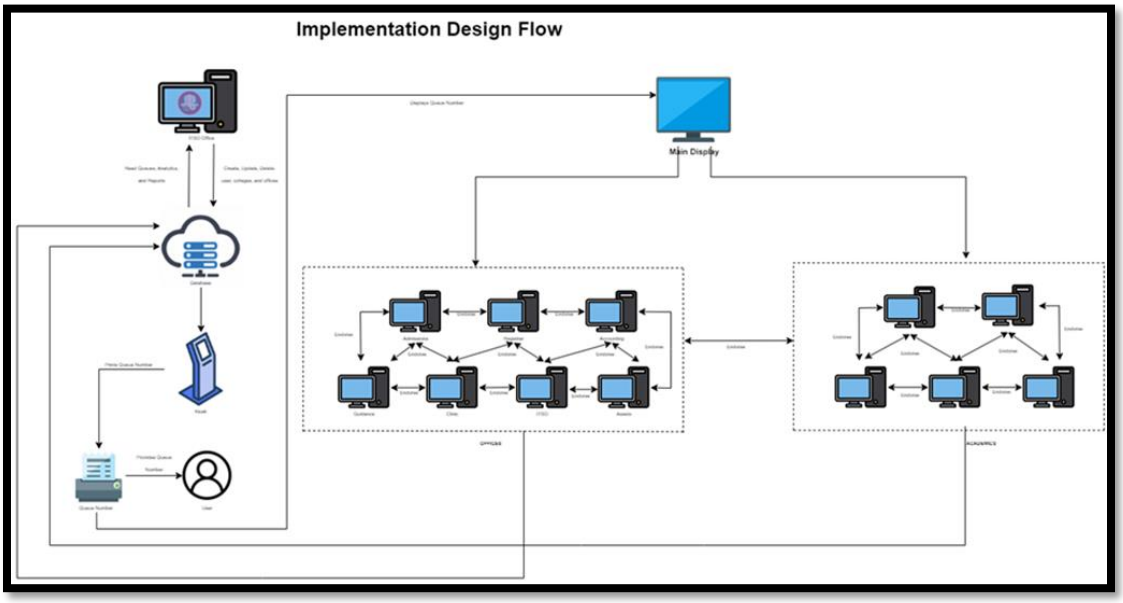
        - One will display the main queue and the other for Academics.
    - Clinic
    - Faculty
    - ITSO

These considerations would greatly benefit students in knowing where they stand in queue.

10. Setting up Academics

- Each College and SHS Program Chairs will be given their own receipt printer for students who opt to be queued for their concerns.
- They will also be given the local host URL to gain access to the system.
- Each College and SHS Program Chairs will also have their own credentials provided by the Super Administrator. These credentials can be used to log in to the system.
- Students will be able to view noticeable queues outside the faculty and near the other offices.

11. Implementation Design Flow



12. Additional Notes

- For the local server to work, the host and all offices must be connected to a single private network.
- The server must be active before starting a transaction.
- Backup the database at least once a week.
- ITSO should reset the queue every day and export the data for reports.
- A monthly maintenance check for all the hardware must be carried out.